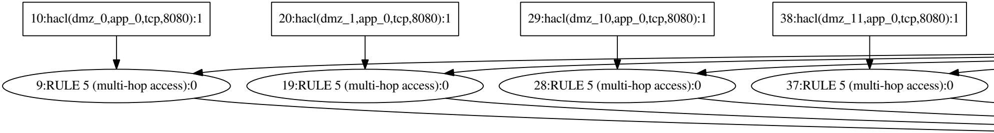
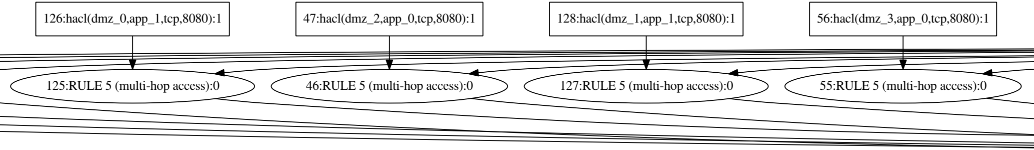
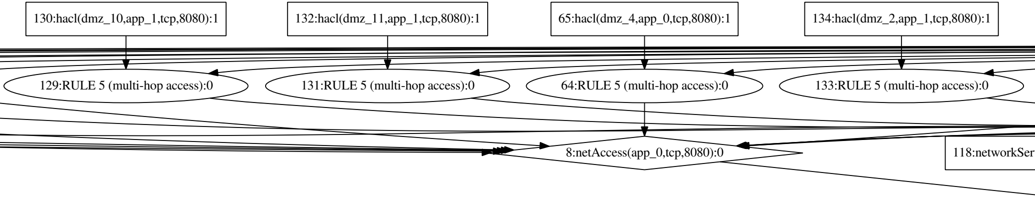
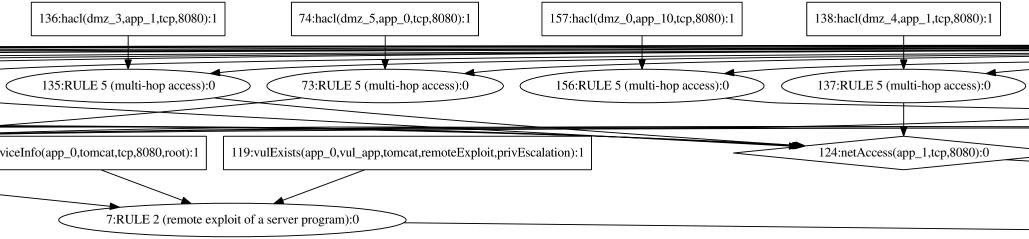
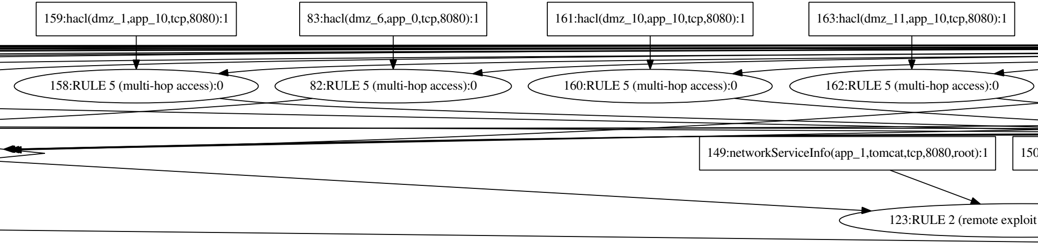
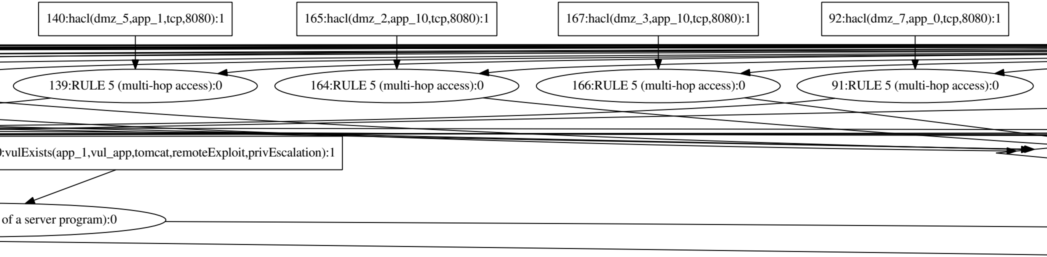

10:hacl(dmz_0,app_0,tcp,8080):1
20:hacl(dmz_1,app_0,tcp,8080):1
29:hacl(dmz_10,app_0,tcp,8080):1
38:hacl(dmz_11,app_0,tcp,8080):1
9:RULE 5 (multi-hop access):0
19:RULE 5 (multi-hop access):0
28:RULE 5 (multi-hop access):0
37:RULE 5 (multi-hop access):0

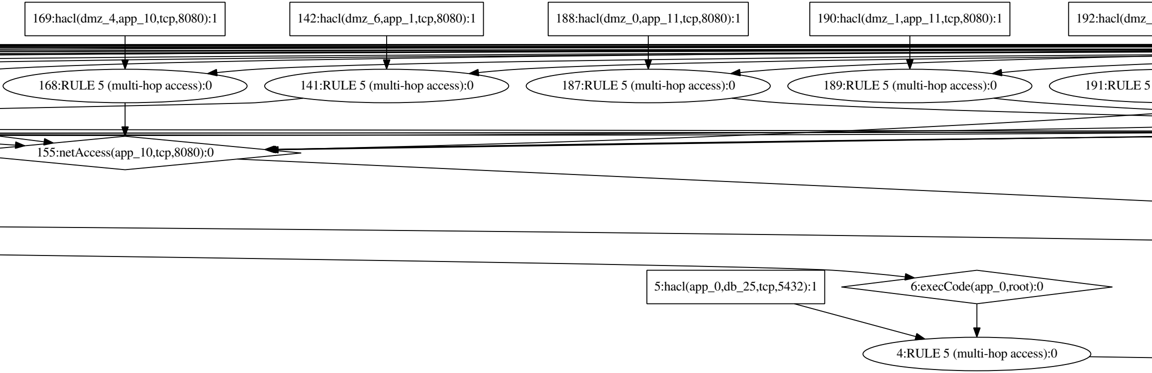

126:hacl(dmz_0,app_1,tcp,8080):1
47:hacl(dmz_2,app_0,tcp,8080):1
128:hacl(dmz_1,app_1,tcp,8080):1
56:hacl(dmz_3,app_0,tcp,8080):1
125:RULE 5 (multi-hop access):0
46:RULE 5 (multi-hop access):0
127:RULE 5 (multi-hop access):0
55:RULE 5 (multi-hop access):0

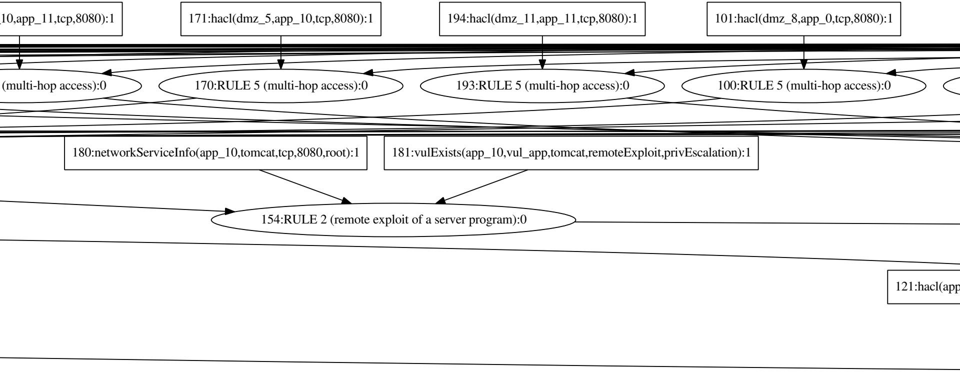

130:hacl(dmz_10,app_1,tcp,8080):1
132:hacl(dmz_11,app_1,tcp,8080):1
65:hacl(dmz_4,app_0,tcp,8080):1
134:hacl(dmz_2,app_1,tcp,8080):1
129:RULE 5 (multi-hop access):0
131:RULE 5 (multi-hop access):0
64:RULE 5 (multi-hop access):0
133:RULE 5 (multi-hop access):0
8:netAccess(app_0,tcp,8080):0
118:networkSer

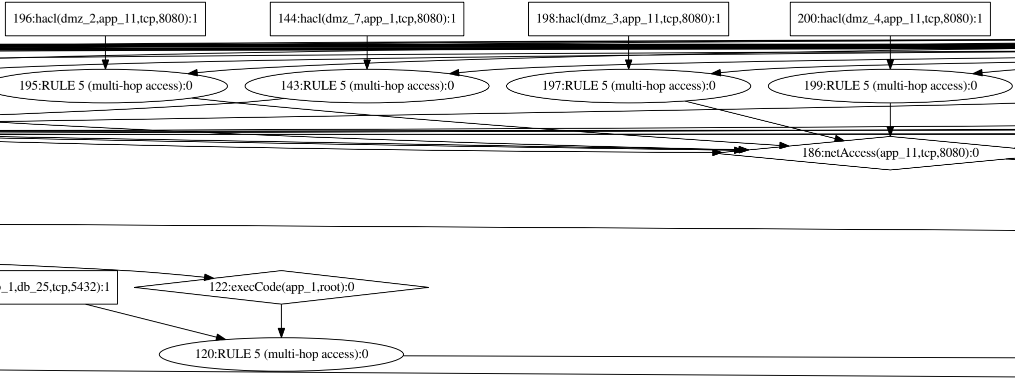

136:hacl(dmz_3,app_1,tcp,8080):1
74:hacl(dmz_5,app_0,tcp,8080):1
157:hacl(dmz_0,app_10,tcp,8080):1
138:hacl(dmz_4,app_1,tcp,8080):1
135:RULE 5 (multi-hop access):0
73:RULE 5 (multi-hop access):0
156:RULE 5 (multi-hop access):0
137:RULE 5 (multi-hop access):0
viceInfo(app_0,tomcat,tcp,8080,root):1
119:vulExists(app_0,vul_app,tomcat,remoteExploit,privEscalation):1
124:netAccess(app_1,tcp,8080):0
7:RULE 2 (remote exploit of a server program):0

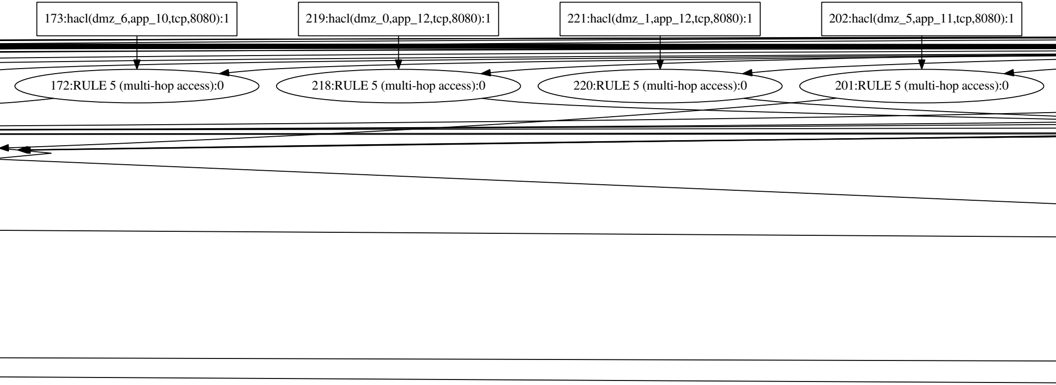

159:hacl(dmz_1,app_10,tcp,8080):1
83:hacl(dmz_6,app_0,tcp,8080):1
161:hacl(dmz_10,app_10,tcp,8080):1
163:hacl(dmz_11,app_10,tcp,8080):1
158:RULE 5 (multi-hop access):0
82:RULE 5 (multi-hop access):0
160:RULE 5 (multi-hop access):0
162:RULE 5 (multi-hop access):0
149:networkServiceInfo(app_1,tomcat,tcp,8080,root):1
150
123:RULE 2 (remote exploit

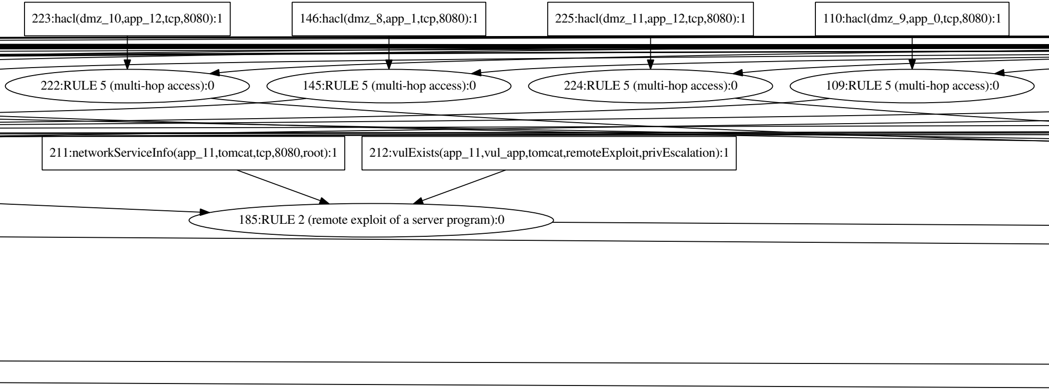

140:hacl(dmz_5,app_1,tcp,8080):1
165:hacl(dmz_2,app_10,tcp,8080):1
167:hacl(dmz_3,app_10,tcp,8080):1
92:hacl(dmz_7,app_0,tcp,8080):1
139:RULE 5 (multi-hop access):0
164:RULE 5 (multi-hop access):0
166:RULE 5 (multi-hop access):0
91:RULE 5 (multi-hop access):0
0:vulExists(app_1,vul_app,tomcat,remoteExploit,privEscalation):1
of a server program):0

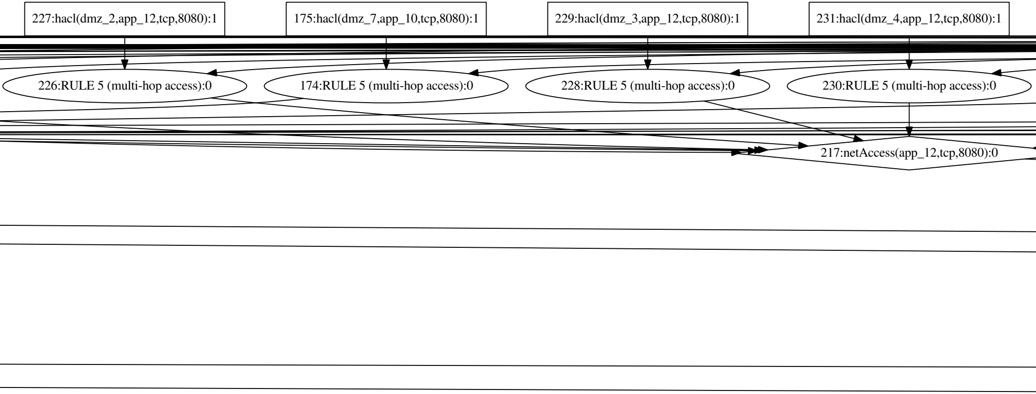

169:hacl(dmz_4,app_10,tcp,8080):1
142:hacl(dmz_6,app_1,tcp,8080):1
188:hacl(dmz_0,app_11,tcp,8080):1
190:hacl(dmz_1,app_11,tcp,8080):1
192:hacl(dmz_
168:RULE 5 (multi-hop access):0
141:RULE 5 (multi-hop access):0
187:RULE 5 (multi-hop access):0
189:RULE 5 (multi-hop access):0
191:RULE 5
155:netAccess(app_10,tcp,8080):0
5:hacl(app_0,db_25,tcp,5432):1
6:execCode(app_0,root):0
4:RULE 5 (multi-hop access):0

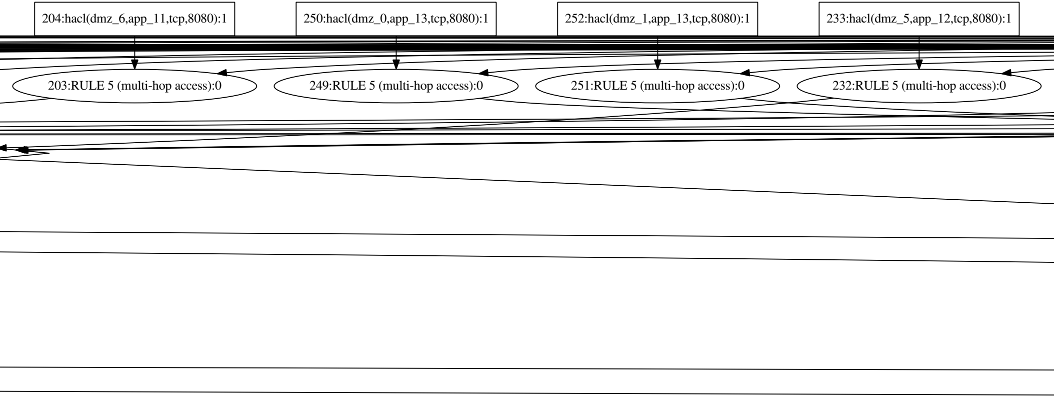

10,app_11,tcp,8080):1
171:hacl(dmz_5,app_10,tcp,8080):1
194:hacl(dmz_11,app_11,tcp,8080):1
101:hacl(dmz_8,app_0,tcp,8080):1
(multi-hop access):0
170:RULE 5 (multi-hop access):0
193:RULE 5 (multi-hop access):0
100:RULE 5 (multi-hop access):0
180:networkServiceInfo(app_10,tomcat,tcp,8080,root):1
181:vulExists(app_10,vul_app,tomcat,remoteExploit,privEscalation):1
154:RULE 2 (remote exploit of a server program):0
121:hacl(app

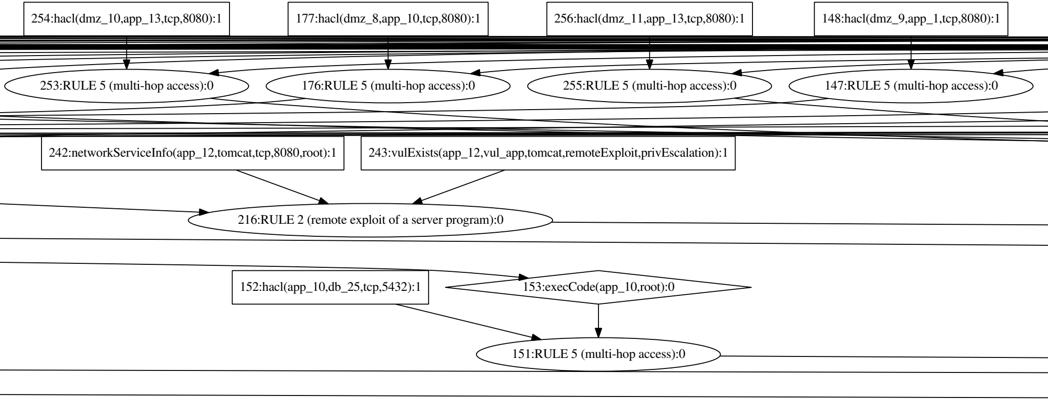

196:hacl(dmz_2,app_11,tcp,8080):1
144:hacl(dmz_7,app_1,tcp,8080):1
198:hacl(dmz_3,app_11,tcp,8080):1
200:hacl(dmz_4,app_11,tcp,8080):1
195:RULE 5 (multi-hop access):0
143:RULE 5 (multi-hop access):0
197:RULE 5 (multi-hop access):0
199:RULE 5 (multi-hop access):0
186:netAccess(app_11,tcp,8080):0
_1,db_25,tcp,5432):1
122:execCode(app_1,root):0
120:RULE 5 (multi-hop access):0

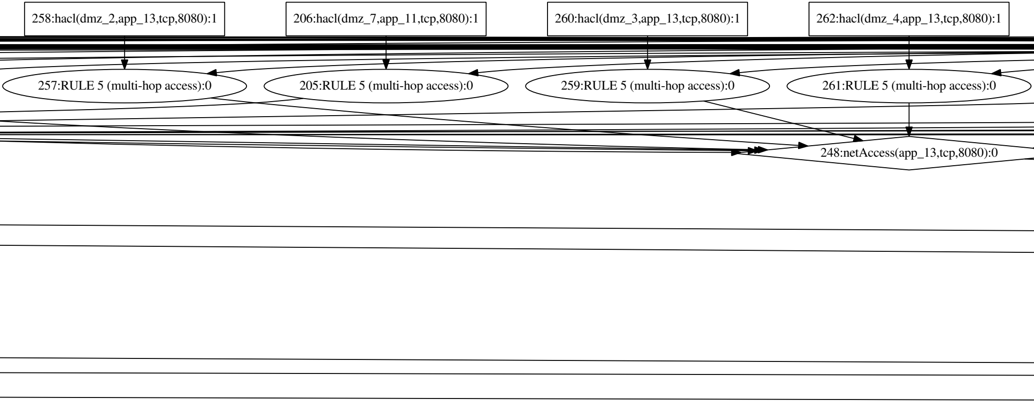

173:hacl(dmz_6,app_10,tcp,8080):1
219:hacl(dmz_0,app_12,tcp,8080):1
221:hacl(dmz_1,app_12,tcp,8080):1
202:hacl(dmz_5,app_11,tcp,8080):1
172:RULE 5 (multi-hop access):0
218:RULE 5 (multi-hop access):0
220:RULE 5 (multi-hop access):0
201:RULE 5 (multi-hop access):0

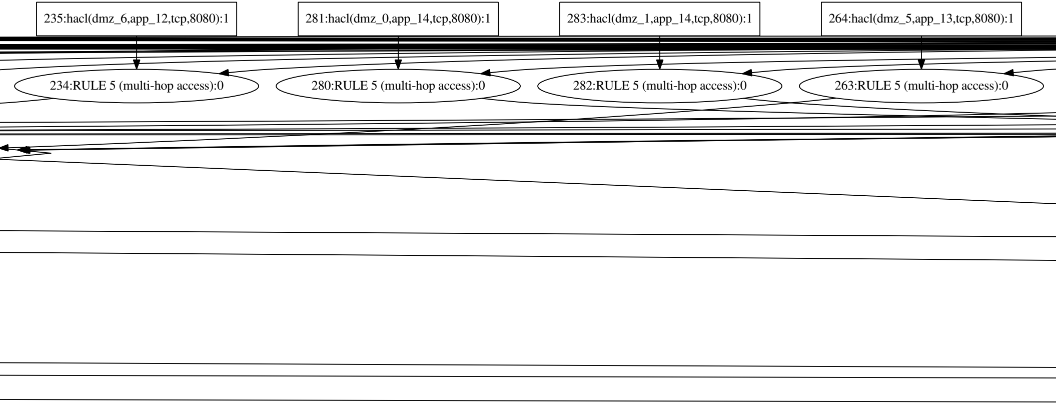

223:hacl(dmz_10,app_12,tcp,8080):1
146:hacl(dmz_8,app_1,tcp,8080):1
225:hacl(dmz_11,app_12,tcp,8080):1
110:hacl(dmz_9,app_0,tcp,8080):1
222:RULE 5 (multi-hop access):0
145:RULE 5 (multi-hop access):0
224:RULE 5 (multi-hop access):0
109:RULE 5 (multi-hop access):0
211:networkServiceInfo(app_11,tomcat,tcp,8080,root):1
212:vulExists(app_11,vul_app,tomcat,remoteExploit,privEscalation):1
185:RULE 2 (remote exploit of a server program):0

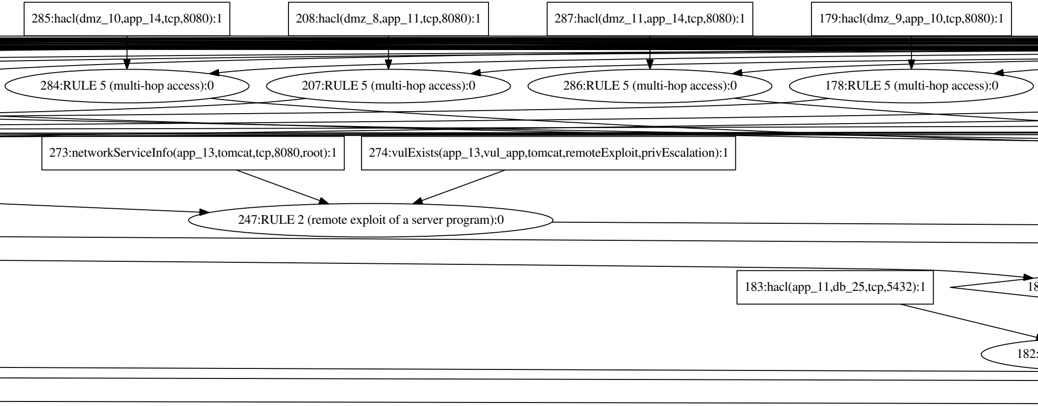

227:hacl(dmz_2,app_12,tcp,8080):1
175:hacl(dmz_7,app_10,tcp,8080):1
229:hacl(dmz_3,app_12,tcp,8080):1
231:hacl(dmz_4,app_12,tcp,8080):1
226:RULE 5 (multi-hop access):0
174:RULE 5 (multi-hop access):0
228:RULE 5 (multi-hop access):0
230:RULE 5 (multi-hop access):0
217:netAccess(app_12,tcp,8080):0

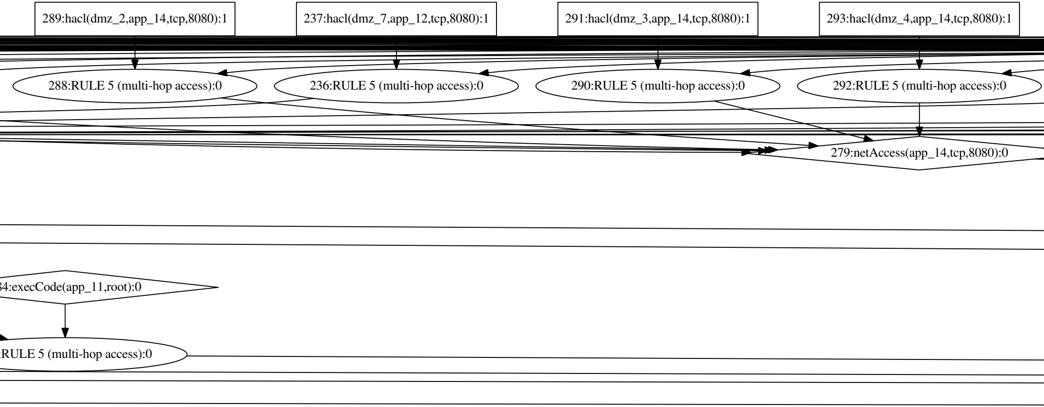

204:hacl(dmz_6,app_11,tcp,8080):1
250:hacl(dmz_0,app_13,tcp,8080):1
252:hacl(dmz_1,app_13,tcp,8080):1
233:hacl(dmz_5,app_12,tcp,8080):1
203:RULE 5 (multi-hop access):0
249:RULE 5 (multi-hop access):0
251:RULE 5 (multi-hop access):0
232:RULE 5 (multi-hop access):0

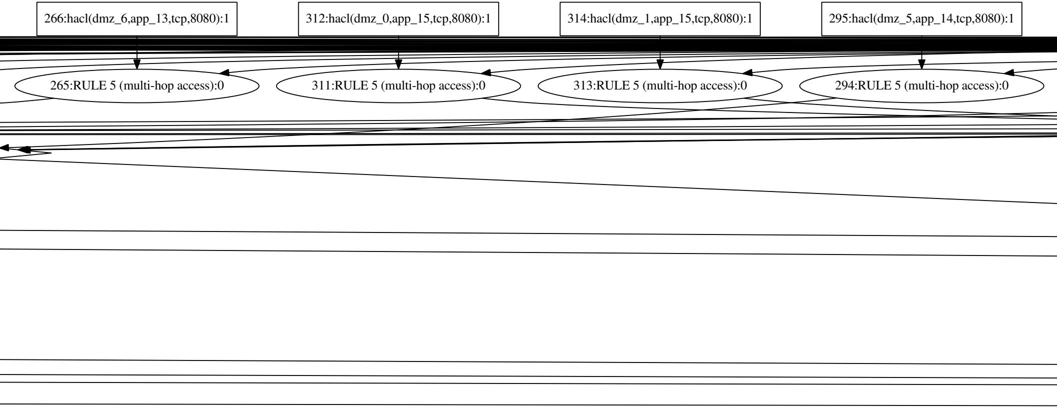

254:hacl(dmz_10,app_13,tcp,8080):1
177:hacl(dmz_8,app_10,tcp,8080):1
256:hacl(dmz_11,app_13,tcp,8080):1
148:hacl(dmz_9,app_1,tcp,8080):1
253:RULE 5 (multi-hop access):0
176:RULE 5 (multi-hop access):0
255:RULE 5 (multi-hop access):0
147:RULE 5 (multi-hop access):0
242:networkServiceInfo(app_12,tomcat,tcp,8080,root):1
243:vulExists(app_12,vul_app,tomcat,remoteExploit,privEscalation):1
216:RULE 2 (remote exploit of a server program):0
152:hacl(app_10,db_25,tcp,5432):1
153:execCode(app_10,root):0
151:RULE 5 (multi-hop access):0

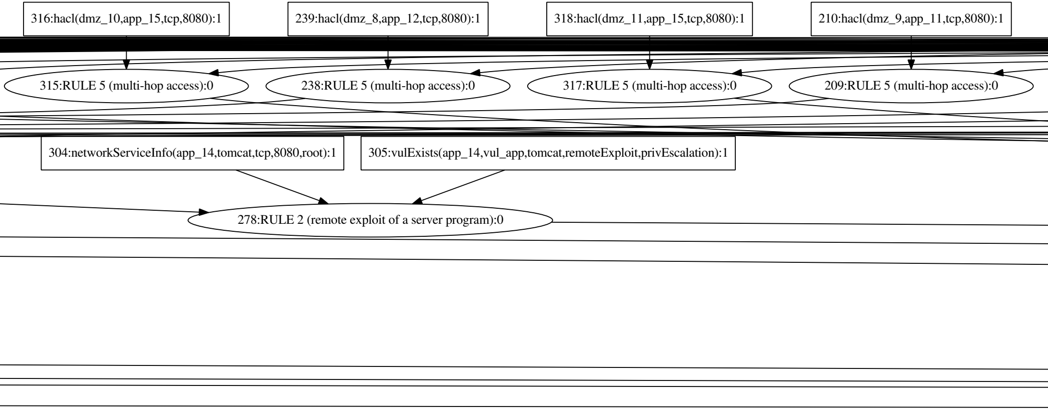

258:hacl(dmz_2,app_13,tcp,8080):1
206:hacl(dmz_7,app_11,tcp,8080):1
260:hacl(dmz_3,app_13,tcp,8080):1
262:hacl(dmz_4,app_13,tcp,8080):1
257:RULE 5 (multi-hop access):0
205:RULE 5 (multi-hop access):0
259:RULE 5 (multi-hop access):0
261:RULE 5 (multi-hop access):0
248:netAccess(app_13,tcp,8080):0

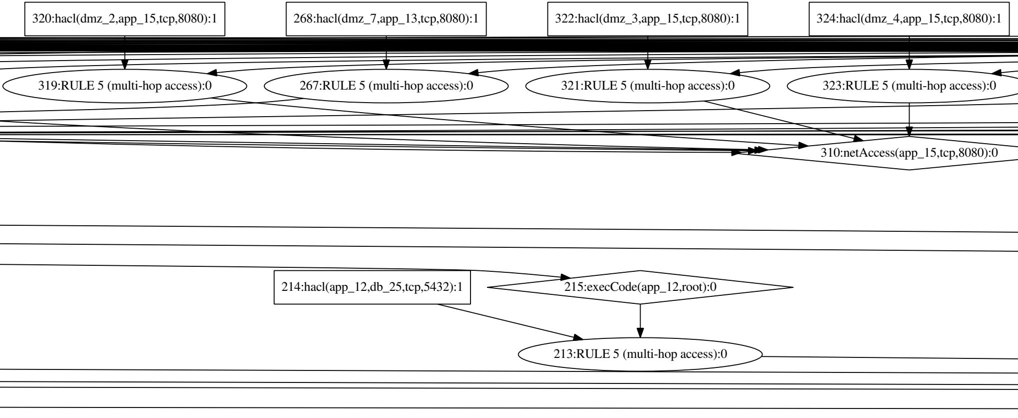

235:hacl(dmz_6,app_12,tcp,8080):1
281:hacl(dmz_0,app_14,tcp,8080):1
283:hacl(dmz_1,app_14,tcp,8080):1
264:hacl(dmz_5,app_13,tcp,8080):1
234:RULE 5 (multi-hop access):0
280:RULE 5 (multi-hop access):0
282:RULE 5 (multi-hop access):0
263:RULE 5 (multi-hop access):0

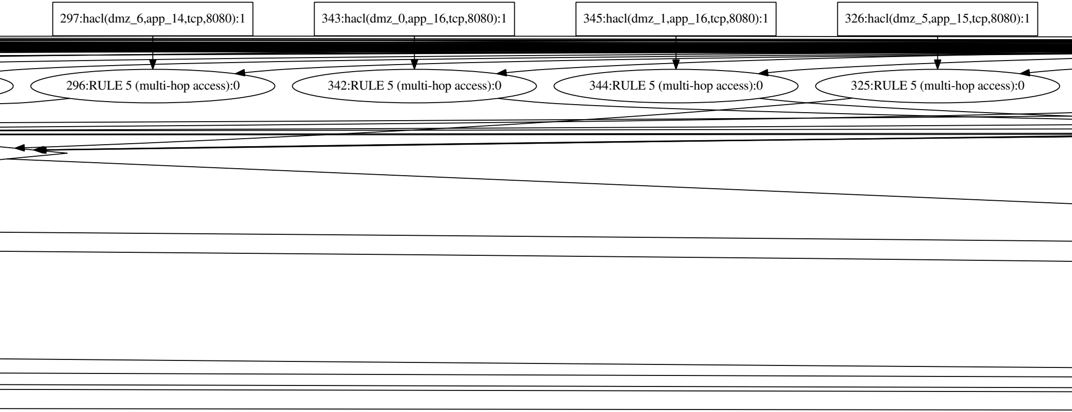

285:hacl(dmz_10,app_14,tcp,8080):1
208:hacl(dmz_8,app_11,tcp,8080):1
287:hacl(dmz_11,app_14,tcp,8080):1
179:hacl(dmz_9,app_10,tcp,8080):1
284:RULE 5 (multi-hop access):0
207:RULE 5 (multi-hop access):0
286:RULE 5 (multi-hop access):0
178:RULE 5 (multi-hop access):0
273:networkServiceInfo(app_13,tomcat,tcp,8080,root):1
274:vulExists(app_13,vul_app,tomcat,remoteExploit,privEscalation):1
247:RULE 2 (remote exploit of a server program):0
183:hacl(app_11,db_25,tcp,5432):1
18
182:

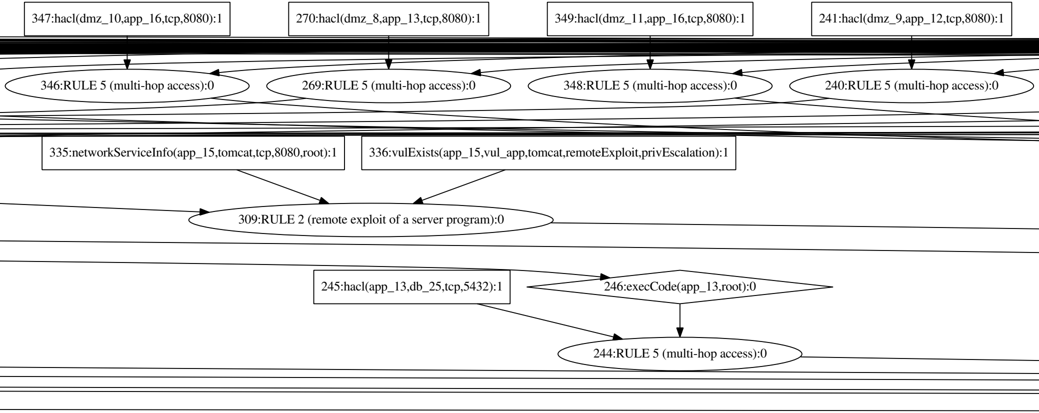

289:hacl(dmz_2,app_14,tcp,8080):1
237:hacl(dmz_7,app_12,tcp,8080):1
291:hacl(dmz_3,app_14,tcp,8080):1
293:hacl(dmz_4,app_14,tcp,8080):1
288:RULE 5 (multi-hop access):0
236:RULE 5 (multi-hop access):0
290:RULE 5 (multi-hop access):0
292:RULE 5 (multi-hop access):0
279:netAccess(app_14,tcp,8080):0
34:execCode(app_11,root):0
RULE 5 (multi-hop access):0

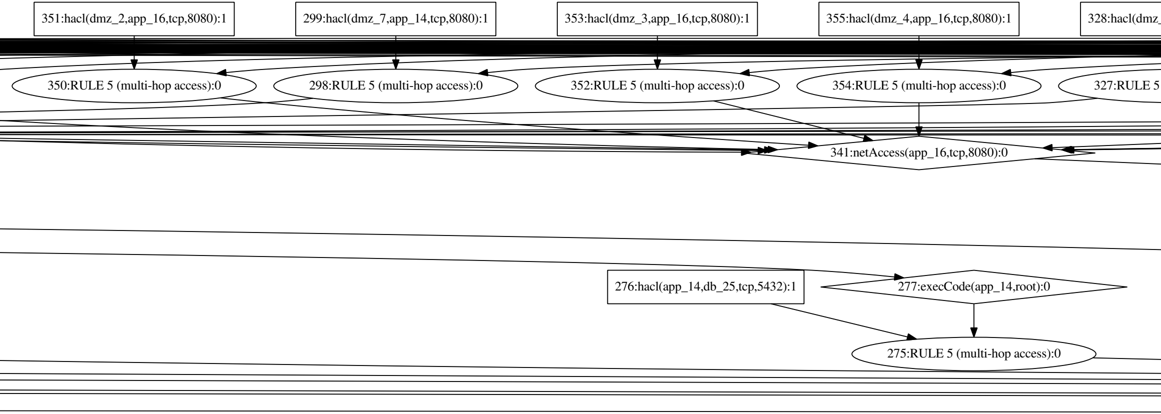

266:hacl(dmz_6,app_13,tcp,8080):1
312:hacl(dmz_0,app_15,tcp,8080):1
314:hacl(dmz_1,app_15,tcp,8080):1
295:hacl(dmz_5,app_14,tcp,8080):1
265:RULE 5 (multi-hop access):0
311:RULE 5 (multi-hop access):0
313:RULE 5 (multi-hop access):0
294:RULE 5 (multi-hop access):0

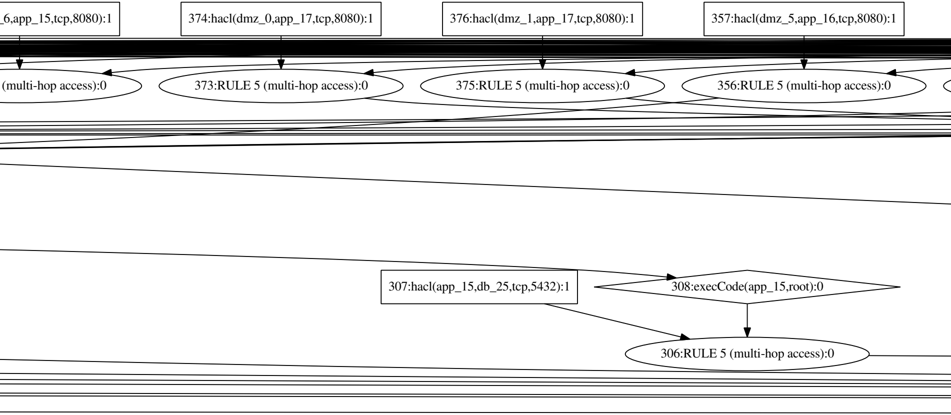

316:hacl(dmz_10,app_15,tcp,8080):1
239:hacl(dmz_8,app_12,tcp,8080):1
318:hacl(dmz_11,app_15,tcp,8080):1
210:hacl(dmz_9,app_11,tcp,8080):1
315:RULE 5 (multi-hop access):0
238:RULE 5 (multi-hop access):0
317:RULE 5 (multi-hop access):0
209:RULE 5 (multi-hop access):0
304:networkServiceInfo(app_14,tomcat,tcp,8080,root):1
305:vulExists(app_14,vul_app,tomcat,remoteExploit,privEscalation):1
278:RULE 2 (remote exploit of a server program):0

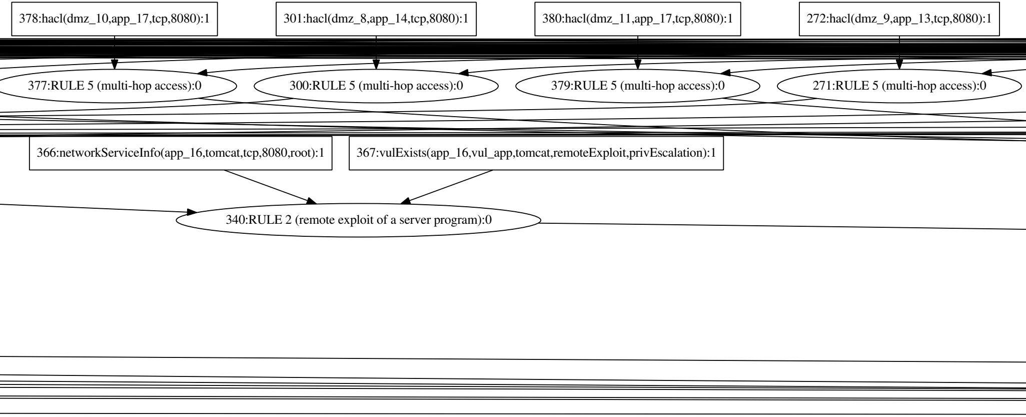

320:hacl(dmz_2,app_15,tcp,8080):1
268:hacl(dmz_7,app_13,tcp,8080):1
322:hacl(dmz_3,app_15,tcp,8080):1
324:hacl(dmz_4,app_15,tcp,8080):1
319:RULE 5 (multi-hop access):0
267:RULE 5 (multi-hop access):0
321:RULE 5 (multi-hop access):0
323:RULE 5 (multi-hop access):0
310:netAccess(app_15,tcp,8080):0
214:hacl(app_12,db_25,tcp,5432):1
215:execCode(app_12,root):0
213:RULE 5 (multi-hop access):0

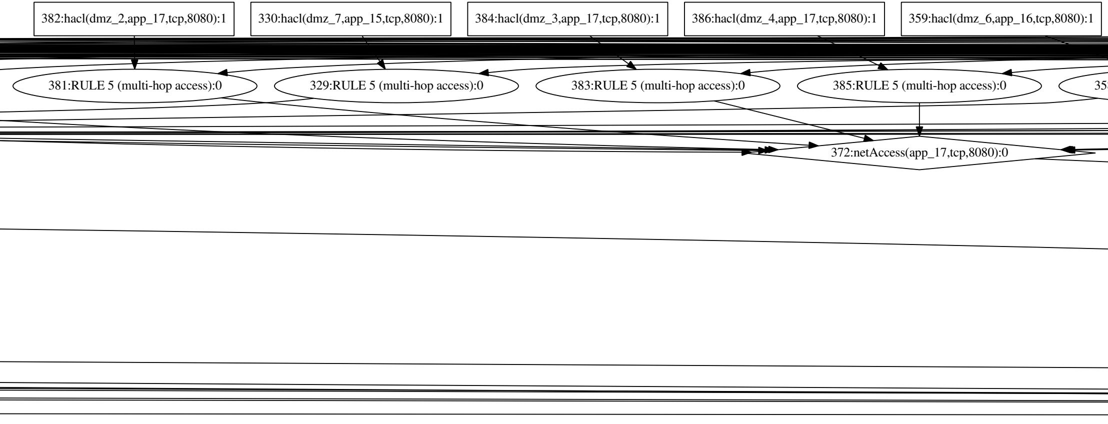

297:hacl(dmz_6,app_14,tcp,8080):1
343:hacl(dmz_0,app_16,tcp,8080):1
345:hacl(dmz_1,app_16,tcp,8080):1
326:hacl(dmz_5,app_15,tcp,8080):1
296:RULE 5 (multi-hop access):0
342:RULE 5 (multi-hop access):0
344:RULE 5 (multi-hop access):0
325:RULE 5 (multi-hop access):0

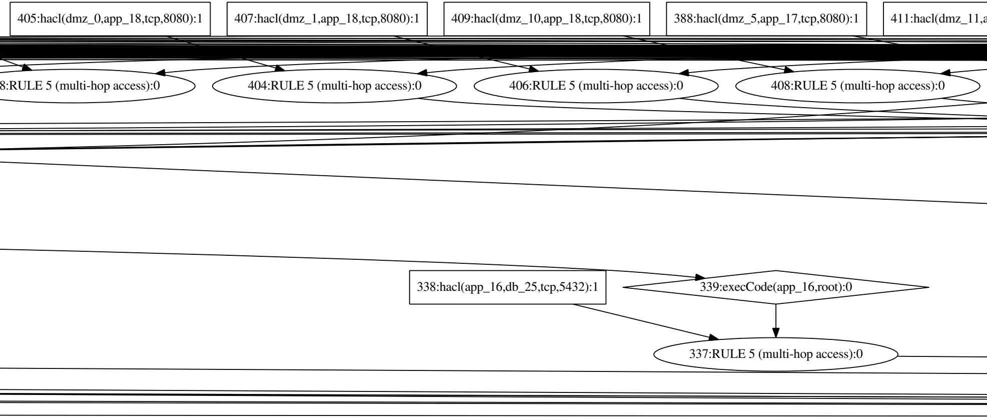

347:hacl(dmz_10,app_16,tcp,8080):1
270:hacl(dmz_8,app_13,tcp,8080):1
349:hacl(dmz_11,app_16,tcp,8080):1
241:hacl(dmz_9,app_12,tcp,8080):1
346:RULE 5 (multi-hop access):0
269:RULE 5 (multi-hop access):0
348:RULE 5 (multi-hop access):0
240:RULE 5 (multi-hop access):0
335:networkServiceInfo(app_15,tomcat,tcp,8080,root):1
336:vulExists(app_15,vul_app,tomcat,remoteExploit,privEscalation):1
309:RULE 2 (remote exploit of a server program):0
245:hacl(app_13,db_25,tcp,5432):1
246:execCode(app_13,root):0
244:RULE 5 (multi-hop access):0

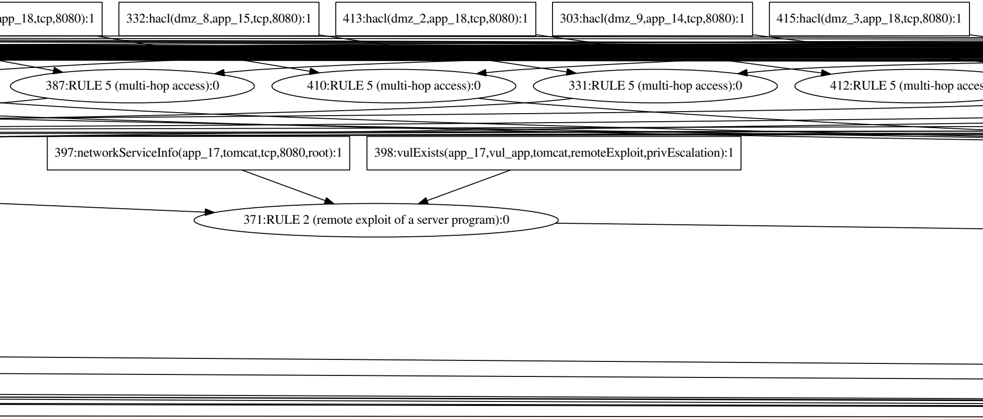

351:hacl(dmz_2,app_16,tcp,8080):1
299:hacl(dmz_7,app_14,tcp,8080):1
353:hacl(dmz_3,app_16,tcp,8080):1
355:hacl(dmz_4,app_16,tcp,8080):1
328:hacl(dmz_
350:RULE 5 (multi-hop access):0
298:RULE 5 (multi-hop access):0
352:RULE 5 (multi-hop access):0
354:RULE 5 (multi-hop access):0
327:RULE 5
341:netAccess(app_16,tcp,8080):0
276:hacl(app_14,db_25,tcp,5432):1
277:execCode(app_14,root):0
275:RULE 5 (multi-hop access):0

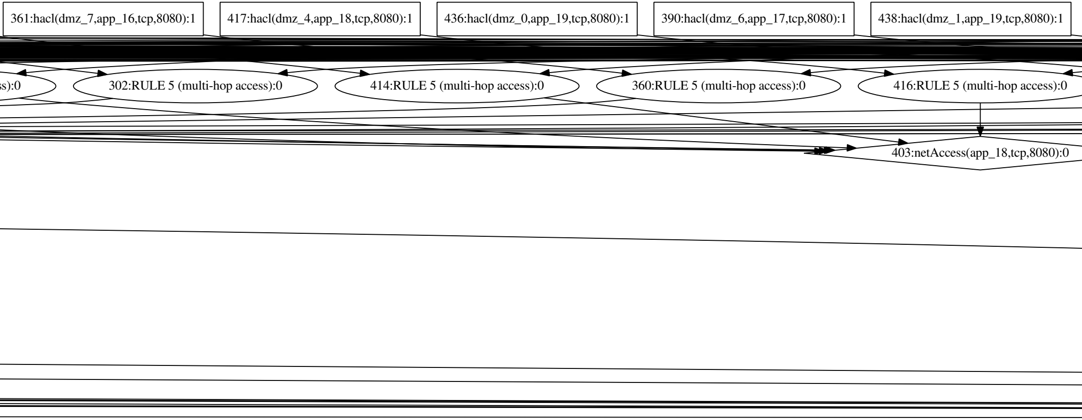

_6,app_15,tcp,8080):1
374:hacl(dmz_0,app_17,tcp,8080):1
376:hacl(dmz_1,app_17,tcp,8080):1
357:hacl(dmz_5,app_16,tcp,8080):1
(multi-hop access):0
373:RULE 5 (multi-hop access):0
375:RULE 5 (multi-hop access):0
356:RULE 5 (multi-hop access):0
307:hacl(app_15,db_25,tcp,5432):1
308:execCode(app_15,root):0
306:RULE 5 (multi-hop access):0

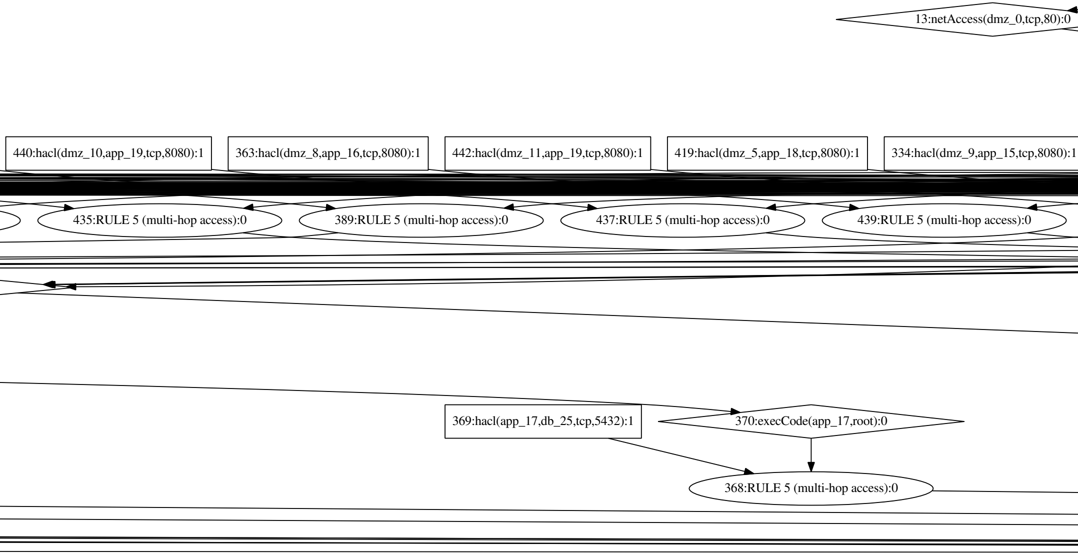

378:hacl(dmz_10,app_17,tcp,8080):1
301:hacl(dmz_8,app_14,tcp,8080):1
380:hacl(dmz_11,app_17,tcp,8080):1
272:hacl(dmz_9,app_13,tcp,8080):1
377:RULE 5 (multi-hop access):0
300:RULE 5 (multi-hop access):0
379:RULE 5 (multi-hop access):0
271:RULE 5 (multi-hop access):0
366:networkServiceInfo(app_16,tomcat,tcp,8080,root):1
367:vulExists(app_16,vul_app,tomcat,remoteExploit,privEscalation):1
340:RULE 2 (remote exploit of a server program):0

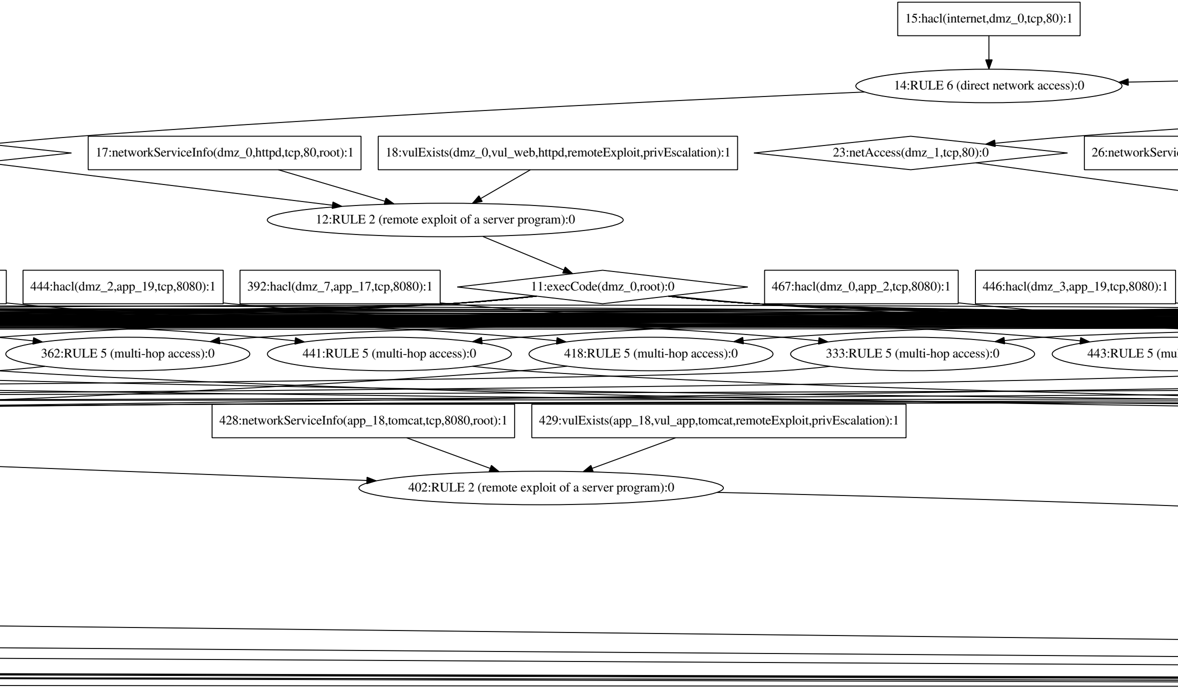

382:hacl(dmz_2,app_17,tcp,8080):1
330:hacl(dmz_7,app_15,tcp,8080):1
384:hacl(dmz_3,app_17,tcp,8080):1
386:hacl(dmz_4,app_17,tcp,8080):1
359:hacl(dmz_6,app_16,tcp,8080):1
381:RULE 5 (multi-hop access):0
329:RULE 5 (multi-hop access):0
383:RULE 5 (multi-hop access):0
385:RULE 5 (multi-hop access):0
372:netAccess(app_17,tcp,8080):0

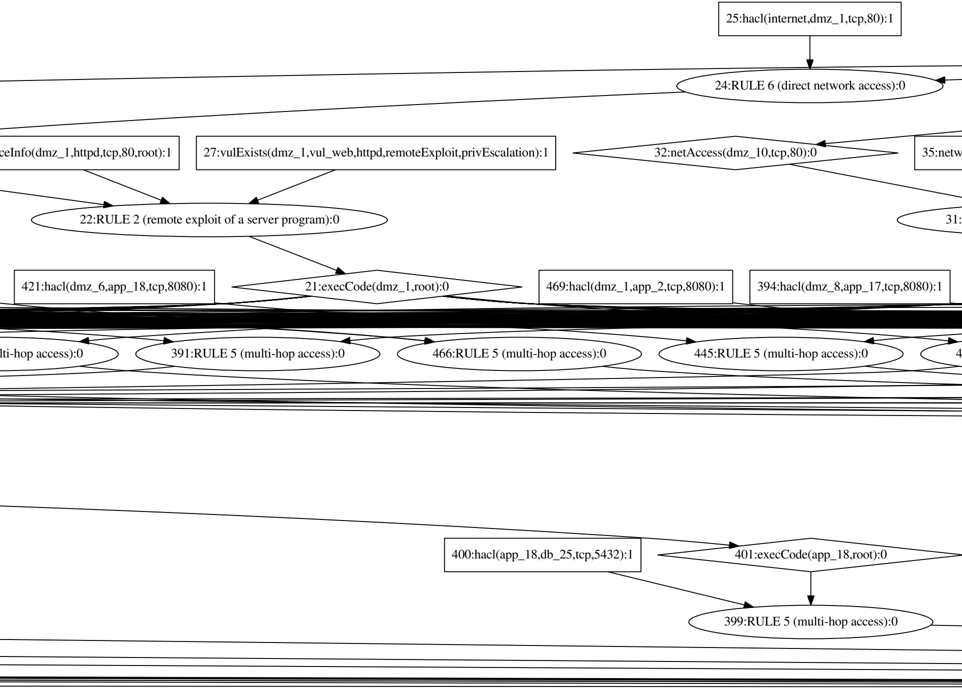

405:hacl(dmz_0,app_18,tcp,8080):1
407:hacl(dmz_1,app_18,tcp,8080):1
409:hacl(dmz_10,app_18,tcp,8080):1
388:hacl(dmz_5,app_17,tcp,8080):1
411:hacl(dmz_11,a
8:RULE 5 (multi-hop access):0
404:RULE 5 (multi-hop access):0
406:RULE 5 (multi-hop access):0
408:RULE 5 (multi-hop access):0
338:hacl(app_16,db_25,tcp,5432):1
339:execCode(app_16,root):0
337:RULE 5 (multi-hop access):0

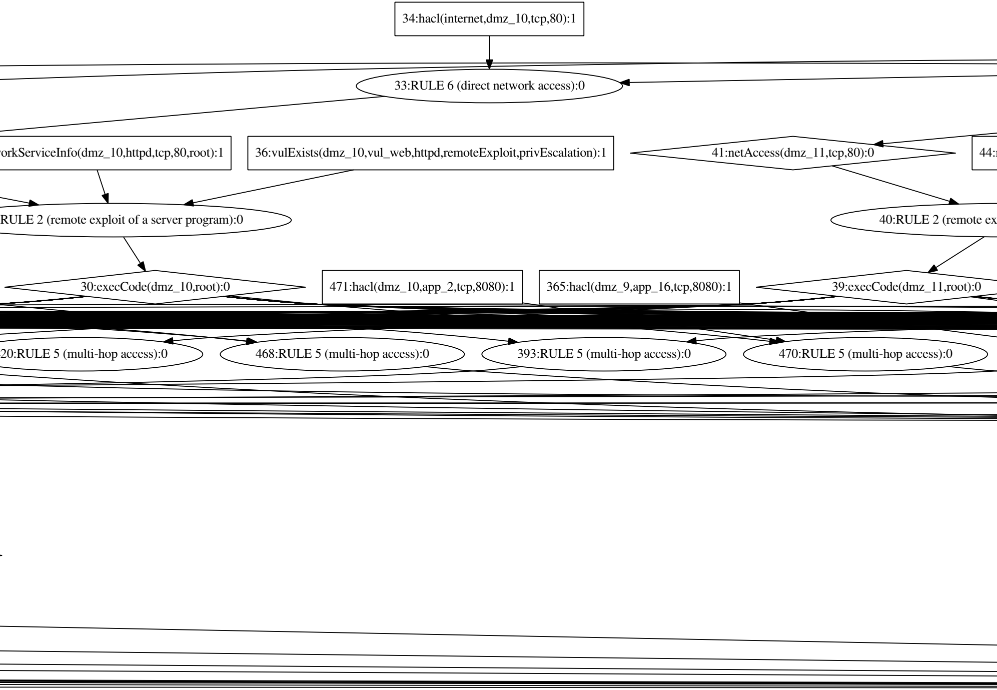

pp_18,tcp,8080):1
332:hacl(dmz_8,app_15,tcp,8080):1
413:hacl(dmz_2,app_18,tcp,8080):1
303:hacl(dmz_9,app_14,tcp,8080):1
415:hacl(dmz_3,app_18,tcp,8080):1
387:RULE 5 (multi-hop access):0
410:RULE 5 (multi-hop access):0
331:RULE 5 (multi-hop access):0
412:RULE 5 (multi-hop acces
397:networkServiceInfo(app_17,tomcat,tcp,8080,root):1
398:vulExists(app_17,vul_app,tomcat,remoteExploit,privEscalation):1
371:RULE 2 (remote exploit of a server program):0

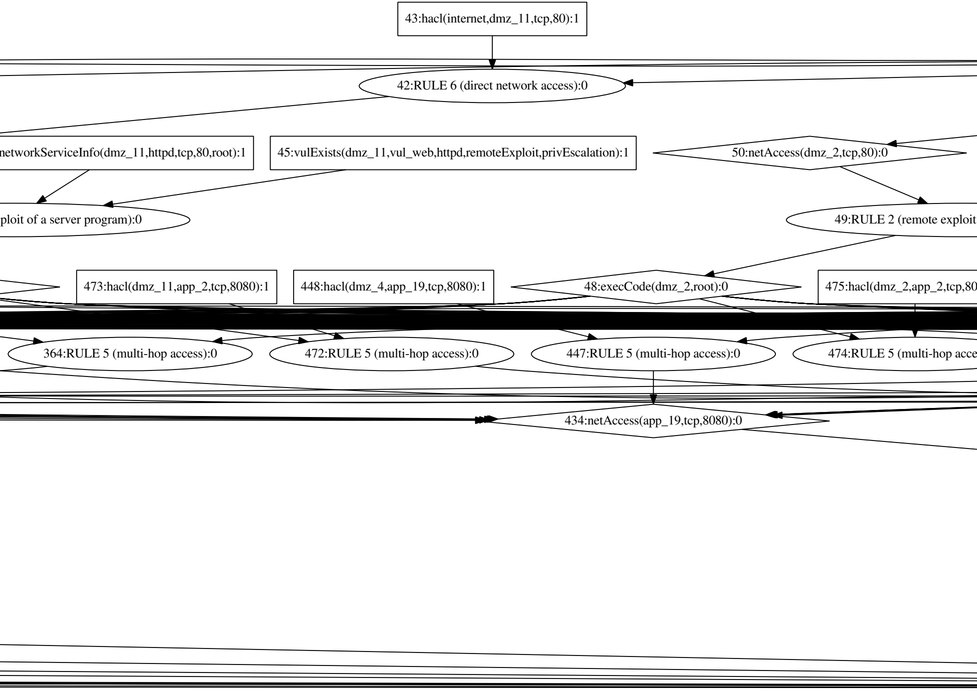

361:hacl(dmz_7,app_16,tcp,8080):1
417:hacl(dmz_4,app_18,tcp,8080):1
436:hacl(dmz_0,app_19,tcp,8080):1
390:hacl(dmz_6,app_17,tcp,8080):1
438:hacl(dmz_1,app_19,tcp,8080):1
ss):0
302:RULE 5 (multi-hop access):0
414:RULE 5 (multi-hop access):0
360:RULE 5 (multi-hop access):0
416:RULE 5 (multi-hop access):0
403:netAccess(app_18,tcp,8080):0

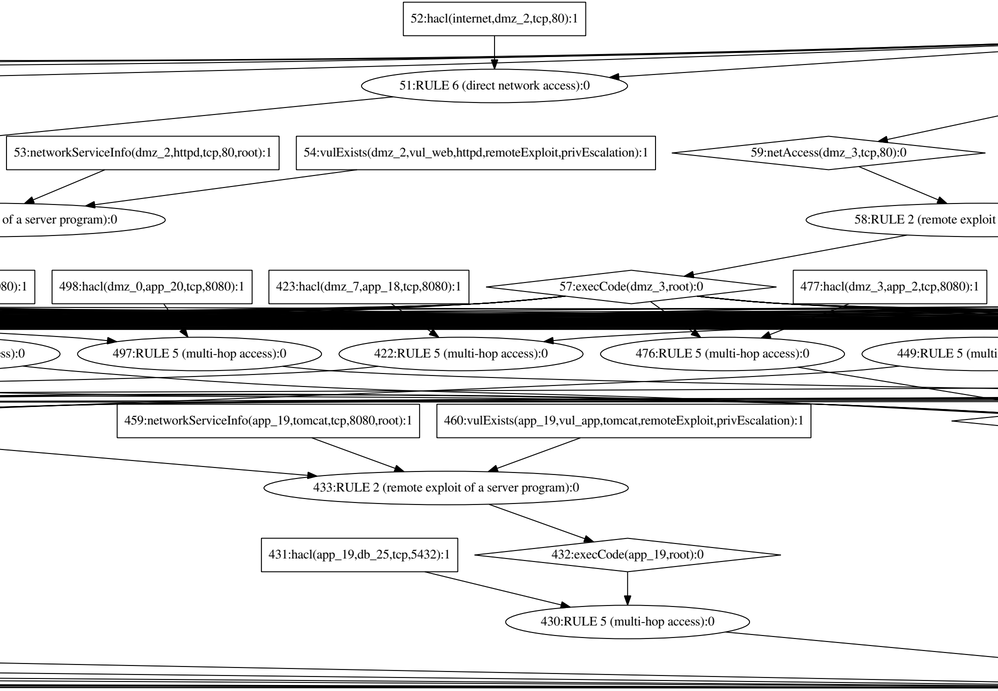

13:netAccess(dmz_0,tcp,80):0
440:hacl(dmz_10,app_19,tcp,8080):1
363:hacl(dmz_8,app_16,tcp,8080):1
442:hacl(dmz_11,app_19,tcp,8080):1
419:hacl(dmz_5,app_18,tcp,8080):1
334:hacl(dmz_9,app_15,tcp,8080):1
435:RULE 5 (multi-hop access):0
389:RULE 5 (multi-hop access):0
437:RULE 5 (multi-hop access):0
439:RULE 5 (multi-hop access):0
369:hacl(app_17,db_25,tcp,5432):1
370:execCode(app_17,root):0
368:RULE 5 (multi-hop access):0

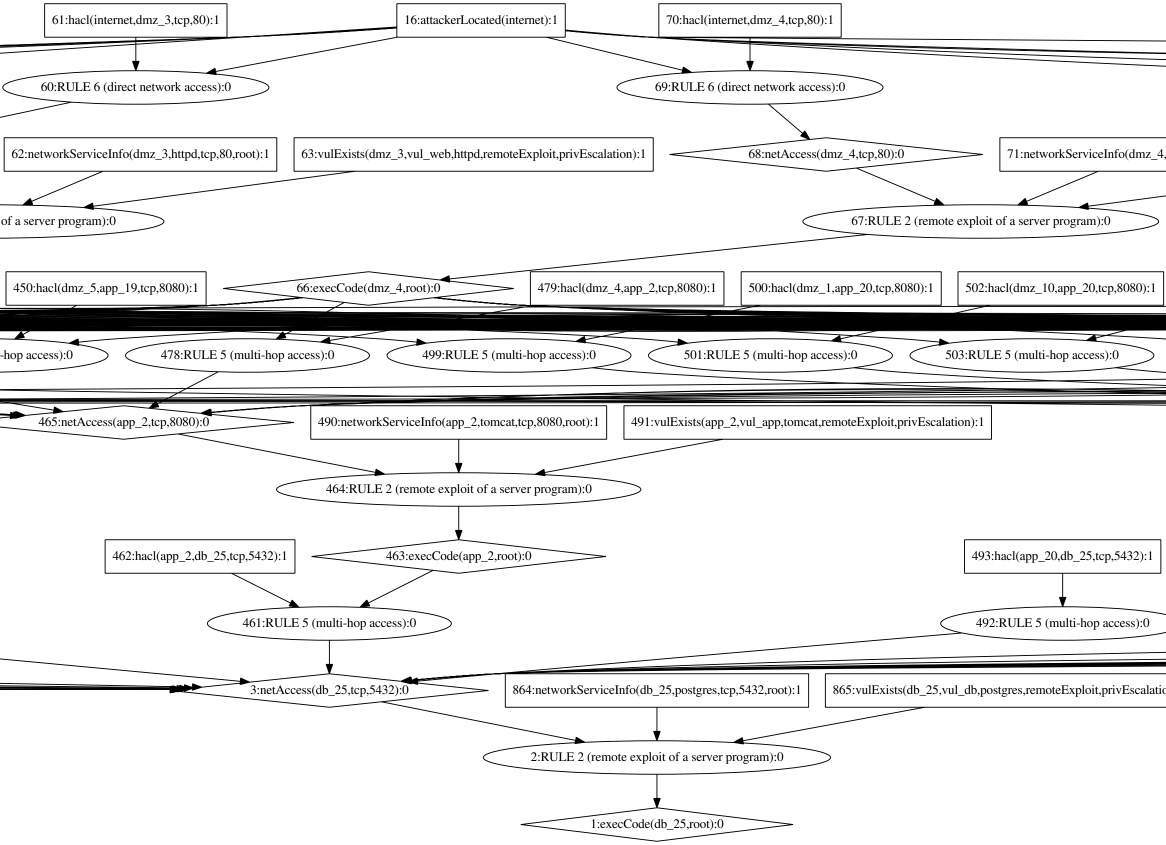

15:hacl(internet,dmz_0,tcp,80):1
14:RULE 6 (direct network access):0
17:networkServiceInfo(dmz_0,httpd,tcp,80,root):1
18:vulExists(dmz_0,vul_web,httpd,remoteExploit,privEscalation):1
23:netAccess(dmz_1,tcp,80):0
26:networkServi
12:RULE 2 (remote exploit of a server program):0
444:hacl(dmz_2,app_19,tcp,8080):1
392:hacl(dmz_7,app_17,tcp,8080):1
11:execCode(dmz_0,root):0
467:hacl(dmz_0,app_2,tcp,8080):1
446:hacl(dmz_3,app_19,tcp,8080):1
362:RULE 5 (multi-hop access):0
441:RULE 5 (multi-hop access):0
418:RULE 5 (multi-hop access):0
333:RULE 5 (multi-hop access):0
443:RULE 5 (mu
428:networkServiceInfo(app_18,tomcat,tcp,8080,root):1
429:vulExists(app_18,vul_app,tomcat,remoteExploit,privEscalation):1
402:RULE 2 (remote exploit of a server program):0

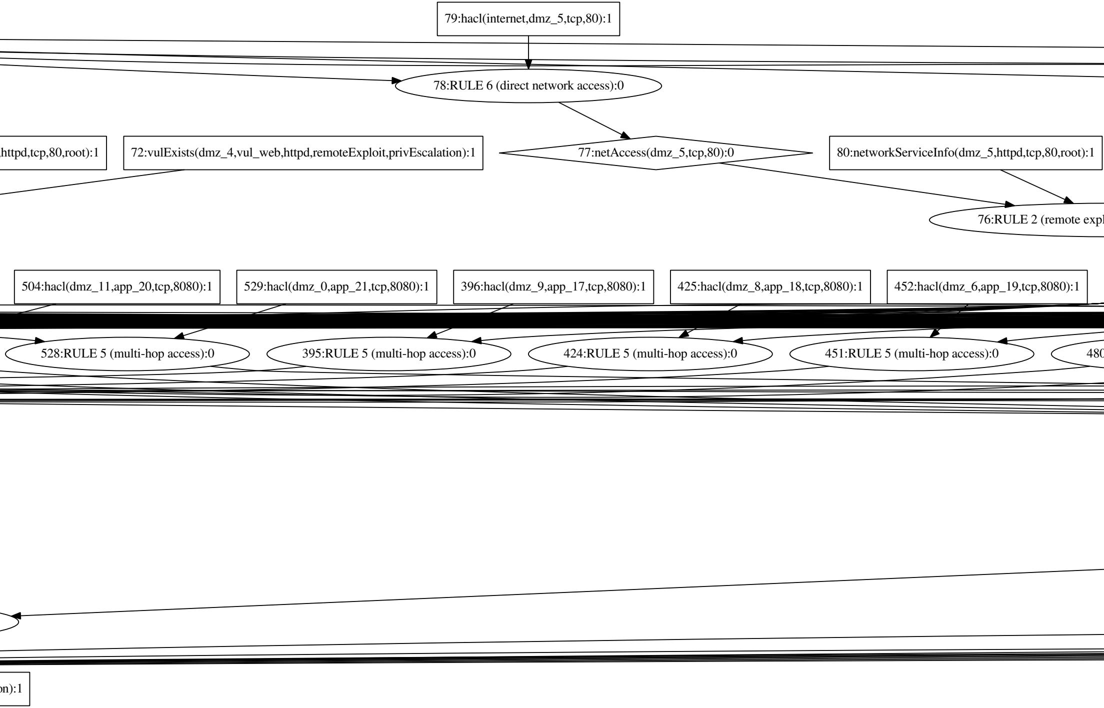

25:hacl(internet,dmz_1,tcp,80):1
24:RULE 6 (direct network access):0
ceInfo(dmz_1,httpd,tcp,80,root):1
27:vulExists(dmz_1,vul_web,httpd,remoteExploit,privEscalation):1
32:netAccess(dmz_10,tcp,80):0
35:netw
22:RULE 2 (remote exploit of a server program):0
31:
421:hacl(dmz_6,app_18,tcp,8080):1
21:execCode(dmz_1,root):0
469:hacl(dmz_1,app_2,tcp,8080):1
394:hacl(dmz_8,app_17,tcp,8080):1
lti-hop access):0
391:RULE 5 (multi-hop access):0
466:RULE 5 (multi-hop access):0
445:RULE 5 (multi-hop access):0
4
400:hacl(app_18,db_25,tcp,5432):1
401:execCode(app_18,root):0
399:RULE 5 (multi-hop access):0

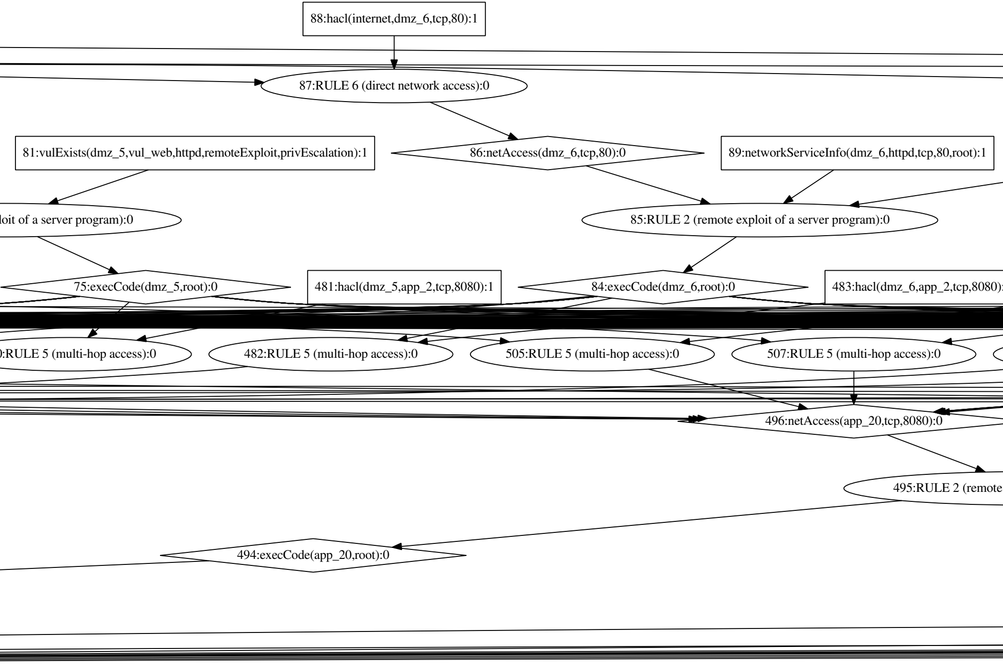

34:hacl(internet,dmz_10,tcp,80):1
33:RULE 6 (direct network access):0
orkServiceInfo(dmz_10,httpd,tcp,80,root):1
36:vulExists(dmz_10,vul_web,httpd,remoteExploit,privEscalation):1
41:netAccess(dmz_11,tcp,80):0
44:
RULE 2 (remote exploit of a server program):0
40:RULE 2 (remote ex
30:execCode(dmz_10,root):0
471:hacl(dmz_10,app_2,tcp,8080):1
365:hacl(dmz_9,app_16,tcp,8080):1
39:execCode(dmz_11,root):0
20:RULE 5 (multi-hop access):0
468:RULE 5 (multi-hop access):0
393:RULE 5 (multi-hop access):0
470:RULE 5 (multi-hop access):0

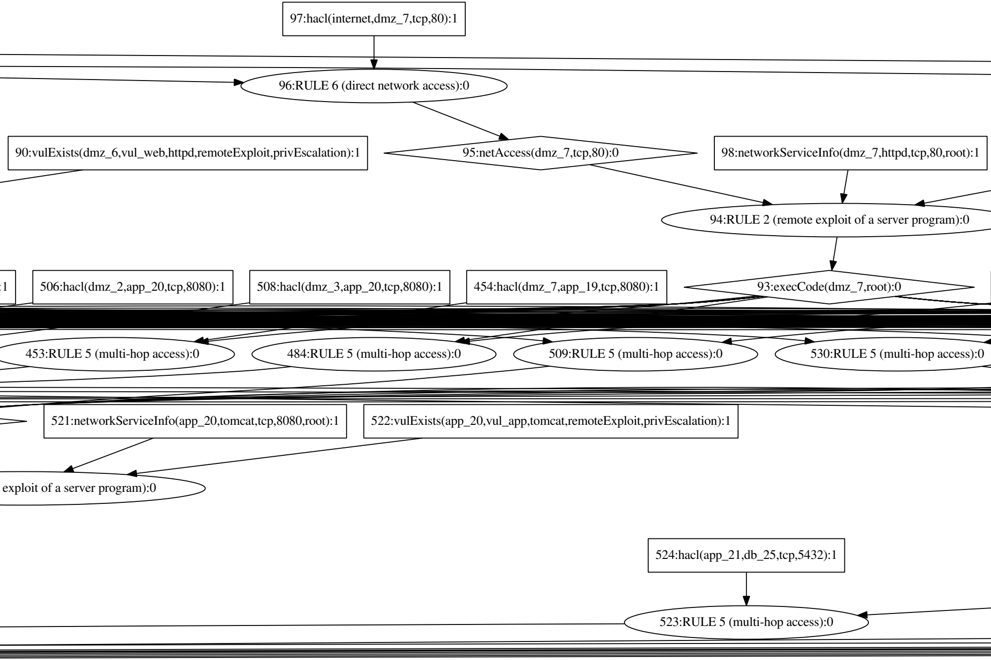

43:hacl(internet,dmz_11,tcp,80):1
42:RULE 6 (direct network access):0
networkServiceInfo(dmz_11,httpd,tcp,80,root):1
45:vulExists(dmz_11,vul_web,httpd,remoteExploit,privEscalation):1
50:netAccess(dmz_2,tcp,80):0
ploit of a server program):0
49:RULE 2 (remote exploit
473:hacl(dmz_11,app_2,tcp,8080):1
448:hacl(dmz_4,app_19,tcp,8080):1
48:execCode(dmz_2,root):0
475:hacl(dmz_2,app_2,tcp,80
364:RULE 5 (multi-hop access):0
472:RULE 5 (multi-hop access):0
447:RULE 5 (multi-hop access):0
474:RULE 5 (multi-hop acce
434:netAccess(app_19,tcp,8080):0

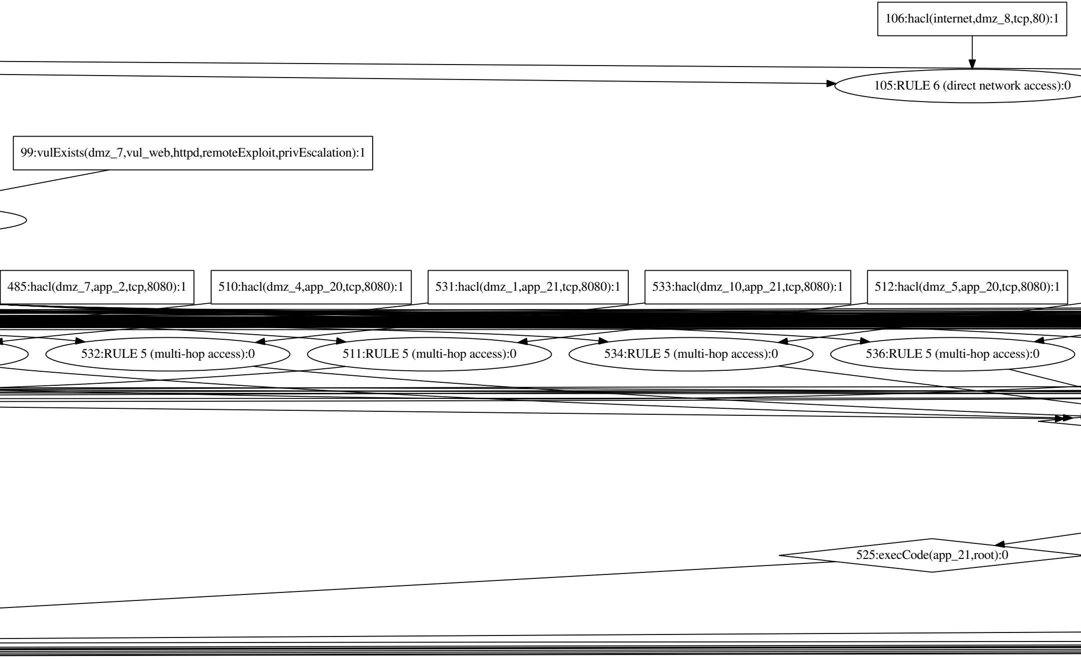

52:hacl(internet,dmz_2,tcp,80):1
51:RULE 6 (direct network access):0
53:networkServiceInfo(dmz_2,httpd,tcp,80,root):1
54:vulExists(dmz_2,vul_web,httpd,remoteExploit,privEscalation):1
59:netAccess(dmz_3,tcp,80):0
of a server program):0
58:RULE 2 (remote exploit
80):1
498:hacl(dmz_0,app_20,tcp,8080):1
423:hacl(dmz_7,app_18,tcp,8080):1
57:execCode(dmz_3,root):0
477:hacl(dmz_3,app_2,tcp,8080):1
ss):0
497:RULE 5 (multi-hop access):0
422:RULE 5 (multi-hop access):0
476:RULE 5 (multi-hop access):0
449:RULE 5 (multi
459:networkServiceInfo(app_19,tomcat,tcp,8080,root):1
460:vulExists(app_19,vul_app,tomcat,remoteExploit,privEscalation):1
433:RULE 2 (remote exploit of a server program):0
431:hacl(app_19,db_25,tcp,5432):1
432:execCode(app_19,root):0
430:RULE 5 (multi-hop access):0

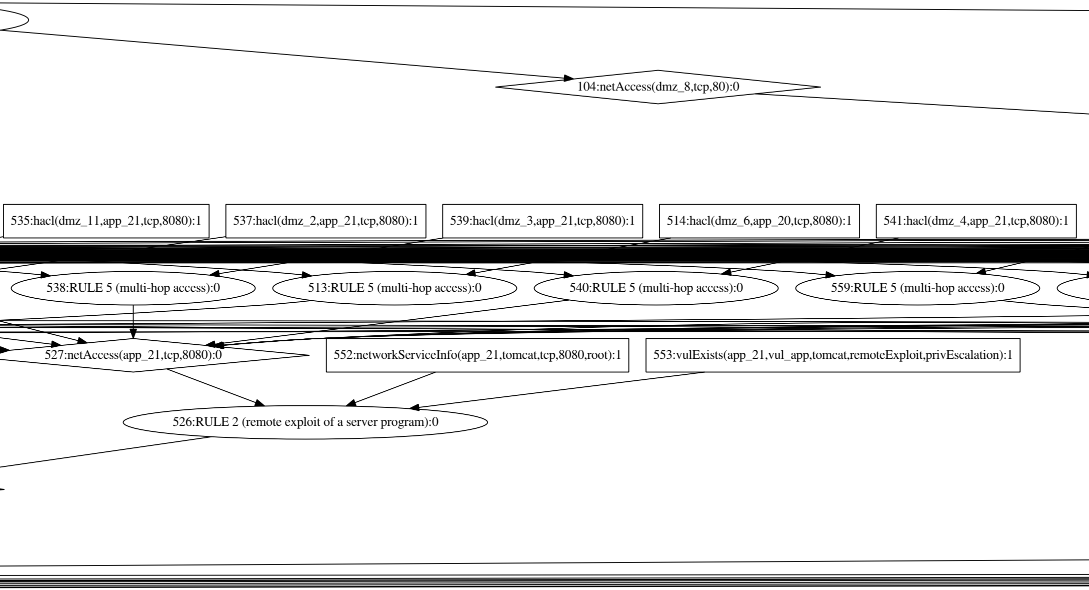

61:hacl(internet,dmz_3,tcp,80):1
16:attackerLocated(internet):1
70:hacl(internet,dmz_4,tcp,80):1
60:RULE 6 (direct network access):0
69:RULE 6 (direct network access):0
62:networkServiceInfo(dmz_3,httpd,tcp,80,root):1
63:vulExists(dmz_3,vul_web,httpd,remoteExploit,privEscalation):1
68:netAccess(dmz_4,tcp,80):0
71:networkServiceInfo(dmz_4,
of a server program):0
67:RULE 2 (remote exploit of a server program):0
450:hacl(dmz_5,app_19,tcp,8080):1
66:execCode(dmz_4,root):0
479:hacl(dmz_4,app_2,tcp,8080):1
500:hacl(dmz_1,app_20,tcp,8080):1
502:hacl(dmz_10,app_20,tcp,8080):1
-hop access):0
478:RULE 5 (multi-hop access):0
499:RULE 5 (multi-hop access):0
501:RULE 5 (multi-hop access):0
503:RULE 5 (multi-hop access):0
465:netAccess(app_2,tcp,8080):0
490:networkServiceInfo(app_2,tomcat,tcp,8080,root):1
491:vulExists(app_2,vul_app,tomcat,remoteExploit,privEscalation):1
464:RULE 2 (remote exploit of a server program):0
462:hacl(app_2,db_25,tcp,5432):1
463:execCode(app_2,root):0
493:hacl(app_20,db_25,tcp,5432):1
461:RULE 5 (multi-hop access):0
492:RULE 5 (multi-hop access):0
3:netAccess(db_25,tcp,5432):0
864:networkServiceInfo(db_25,postgres,tcp,5432,root):1
865:vulExists(db_25,vul_db,postgres,remoteExploit,privEscalatio
2:RULE 2 (remote exploit of a server program):0
1:execCode(db_25,root):0

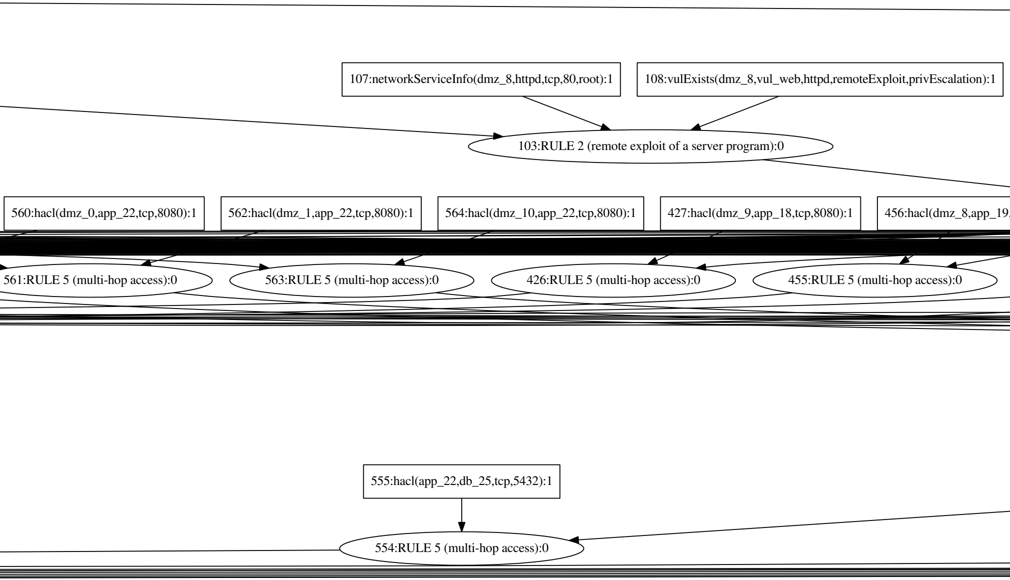

79:hacl(internet,dmz_5,tcp,80):1
78:RULE 6 (direct network access):0
httpd,tcp,80,root):1
72:vulExists(dmz_4,vul_web,httpd,remoteExploit,privEscalation):1
77:netAccess(dmz_5,tcp,80):0
80:networkServiceInfo(dmz_5,httpd,tcp,80,root):1
76:RULE 2 (remote expl
504:hacl(dmz_11,app_20,tcp,8080):1
529:hacl(dmz_0,app_21,tcp,8080):1
396:hacl(dmz_9,app_17,tcp,8080):1
425:hacl(dmz_8,app_18,tcp,8080):1
452:hacl(dmz_6,app_19,tcp,8080):1
528:RULE 5 (multi-hop access):0
395:RULE 5 (multi-hop access):0
424:RULE 5 (multi-hop access):0
451:RULE 5 (multi-hop access):0
480
on):1

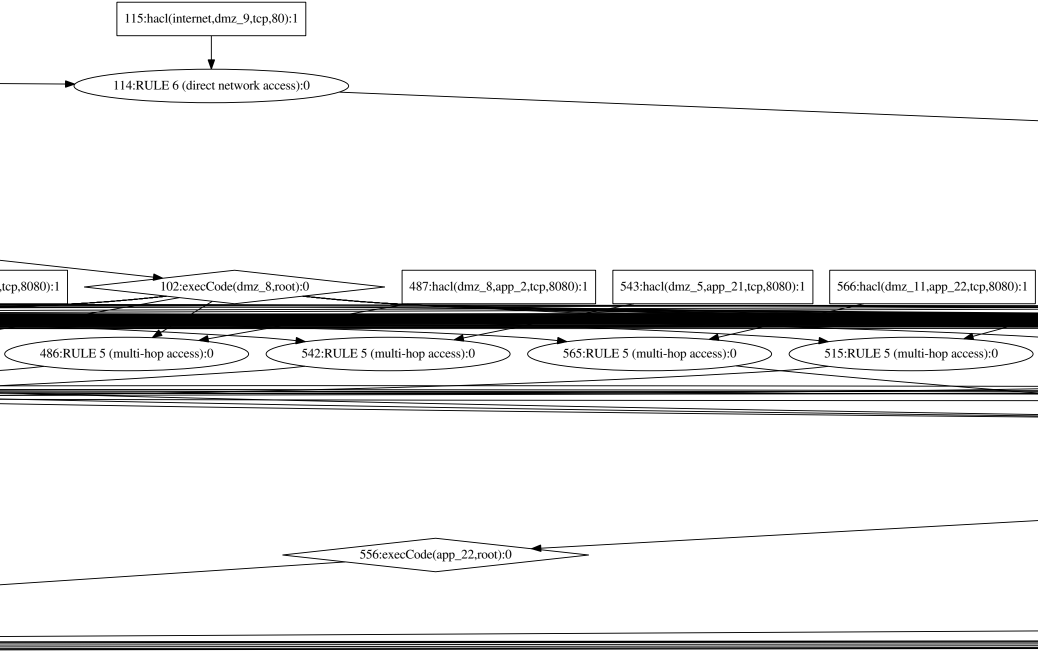

88:hacl(internet,dmz_6,tcp,80):1
87:RULE 6 (direct network access):0
81:vulExists(dmz_5,vul_web,httpd,remoteExploit,privEscalation):1
86:netAccess(dmz_6,tcp,80):0
89:networkServiceInfo(dmz_6,httpd,tcp,80,root):1
loit of a server program):0
85:RULE 2 (remote exploit of a server program):0
75:execCode(dmz_5,root):0
481:hacl(dmz_5,app_2,tcp,8080):1
84:execCode(dmz_6,root):0
483:hacl(dmz_6,app_2,tcp,8080):
:RULE 5 (multi-hop access):0
482:RULE 5 (multi-hop access):0
505:RULE 5 (multi-hop access):0
507:RULE 5 (multi-hop access):0
496:netAccess(app_20,tcp,8080):0
495:RULE 2 (remote
494:execCode(app_20,root):0

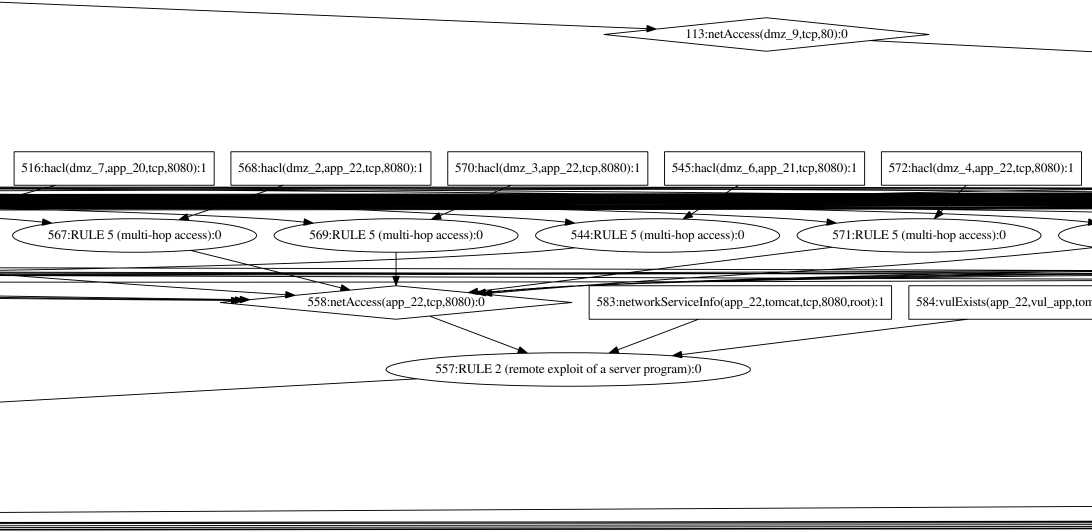

97:hacl(internet,dmz_7,tcp,80):1
96:RULE 6 (direct network access):0
90:vulExists(dmz_6,vul_web,httpd,remoteExploit,privEscalation):1
95:netAccess(dmz_7,tcp,80):0
98:networkServiceInfo(dmz_7,httpd,tcp,80,root):1
94:RULE 2 (remote exploit of a server program):0
:1
506:hacl(dmz_2,app_20,tcp,8080):1
508:hacl(dmz_3,app_20,tcp,8080):1
454:hacl(dmz_7,app_19,tcp,8080):1
93:execCode(dmz_7,root):0
453:RULE 5 (multi-hop access):0
484:RULE 5 (multi-hop access):0
509:RULE 5 (multi-hop access):0
530:RULE 5 (multi-hop access):0
521:networkServiceInfo(app_20,tomcat,tcp,8080,root):1
522:vulExists(app_20,vul_app,tomcat,remoteExploit,privEscalation):1
exploit of a server program):0
524:hacl(app_21,db_25,tcp,5432):1
523:RULE 5 (multi-hop access):0

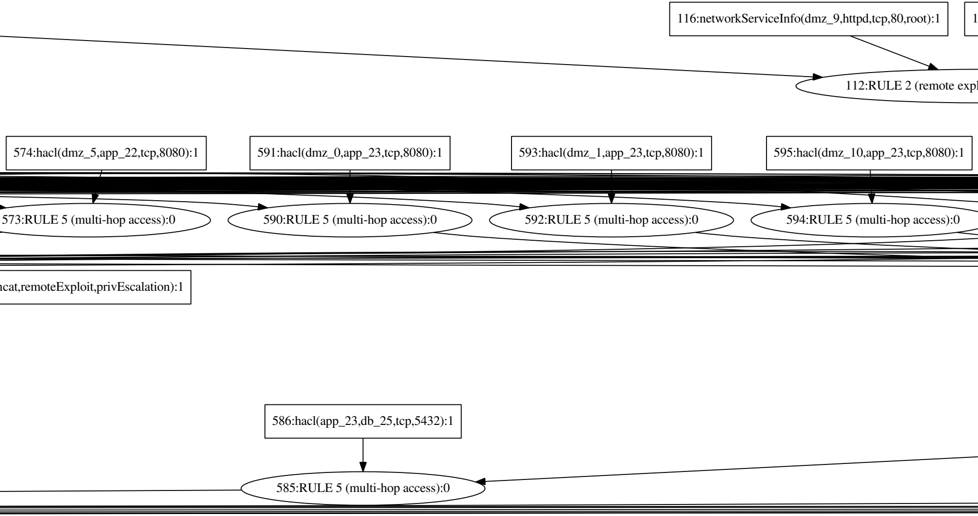

106:hacl(internet,dmz_8,tcp,80):1
105:RULE 6 (direct network access):0
99:vulExists(dmz_7,vul_web,httpd,remoteExploit,privEscalation):1
485:hacl(dmz_7,app_2,tcp,8080):1
510:hacl(dmz_4,app_20,tcp,8080):1
531:hacl(dmz_1,app_21,tcp,8080):1
533:hacl(dmz_10,app_21,tcp,8080):1
512:hacl(dmz_5,app_20,tcp,8080):1
532:RULE 5 (multi-hop access):0
511:RULE 5 (multi-hop access):0
534:RULE 5 (multi-hop access):0
536:RULE 5 (multi-hop access):0
525:execCode(app_21,root):0

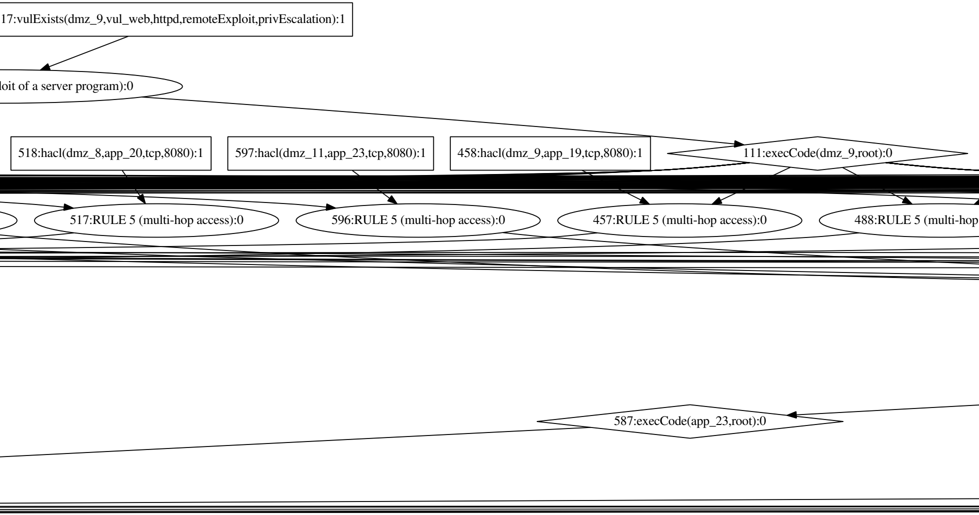

104:netAccess(dmz_8,tcp,80):0
535:hacl(dmz_11,app_21,tcp,8080):1
537:hacl(dmz_2,app_21,tcp,8080):1
539:hacl(dmz_3,app_21,tcp,8080):1
514:hacl(dmz_6,app_20,tcp,8080):1
541:hacl(dmz_4,app_21,tcp,8080):1
538:RULE 5 (multi-hop access):0
513:RULE 5 (multi-hop access):0
540:RULE 5 (multi-hop access):0
559:RULE 5 (multi-hop access):0
527:netAccess(app_21,tcp,8080):0
552:networkServiceInfo(app_21,tomcat,tcp,8080,root):1
553:vulExists(app_21,vul_app,tomcat,remoteExploit,privEscalation):1
526:RULE 2 (remote exploit of a server program):0

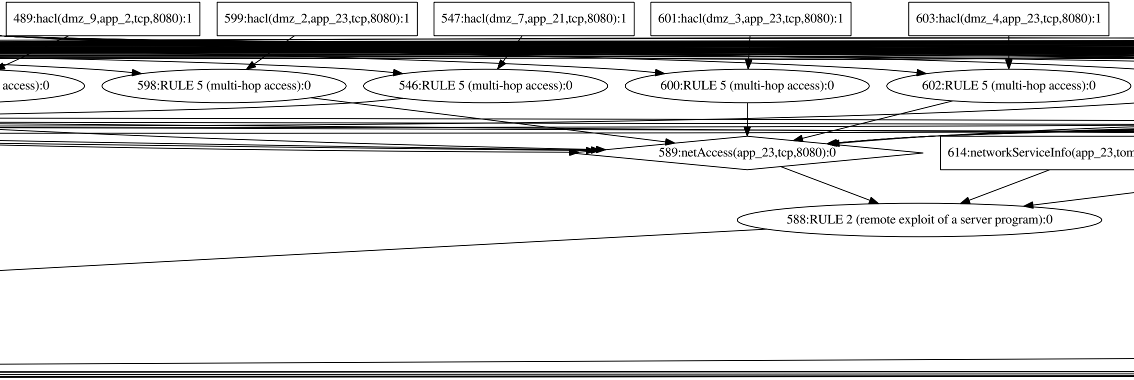

107:networkServiceInfo(dmz_8,httpd,tcp,80,root):1
108:vulExists(dmz_8,vul_web,httpd,remoteExploit,privEscalation):1
103:RULE 2 (remote exploit of a server program):0
560:hacl(dmz_0,app_22,tcp,8080):1
562:hacl(dmz_1,app_22,tcp,8080):1
564:hacl(dmz_10,app_22,tcp,8080):1
427:hacl(dmz_9,app_18,tcp,8080):1
456:hacl(dmz_8,app_19,
561:RULE 5 (multi-hop access):0
563:RULE 5 (multi-hop access):0
426:RULE 5 (multi-hop access):0
455:RULE 5 (multi-hop access):0
555:hacl(app_22,db_25,tcp,5432):1
554:RULE 5 (multi-hop access):0

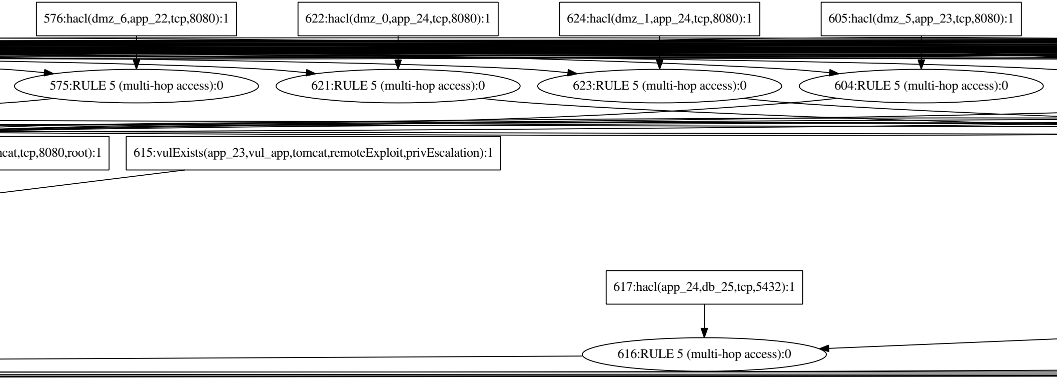

115:hacl(internet,dmz_9,tcp,80):1
114:RULE 6 (direct network access):0
,tcp,8080):1
102:execCode(dmz_8,root):0
487:hacl(dmz_8,app_2,tcp,8080):1
543:hacl(dmz_5,app_21,tcp,8080):1
566:hacl(dmz_11,app_22,tcp,8080):1
486:RULE 5 (multi-hop access):0
542:RULE 5 (multi-hop access):0
565:RULE 5 (multi-hop access):0
515:RULE 5 (multi-hop access):0
556:execCode(app_22,root):0

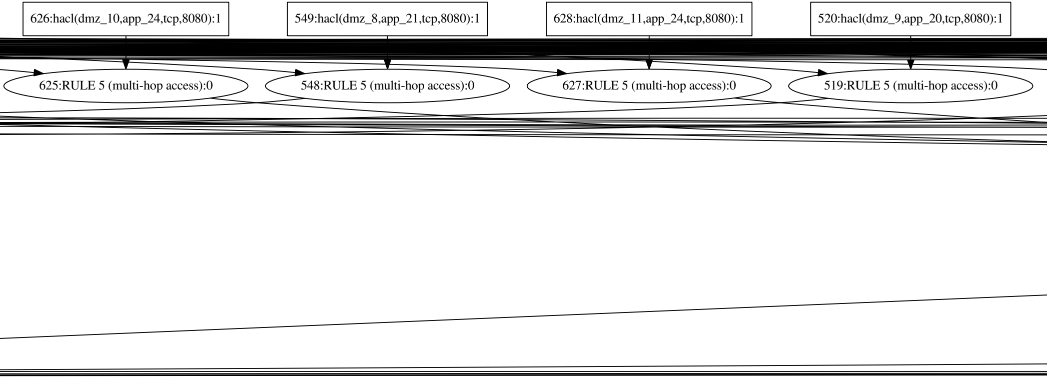

113:netAccess(dmz_9,tcp,80):0
516:hacl(dmz_7,app_20,tcp,8080):1
568:hacl(dmz_2,app_22,tcp,8080):1
570:hacl(dmz_3,app_22,tcp,8080):1
545:hacl(dmz_6,app_21,tcp,8080):1
572:hacl(dmz_4,app_22,tcp,8080):1
567:RULE 5 (multi-hop access):0
569:RULE 5 (multi-hop access):0
544:RULE 5 (multi-hop access):0
571:RULE 5 (multi-hop access):0
558:netAccess(app_22,tcp,8080):0
583:networkServiceInfo(app_22,tomcat,tcp,8080,root):1
584:vulExists(app_22,vul_app,tom
557:RULE 2 (remote exploit of a server program):0

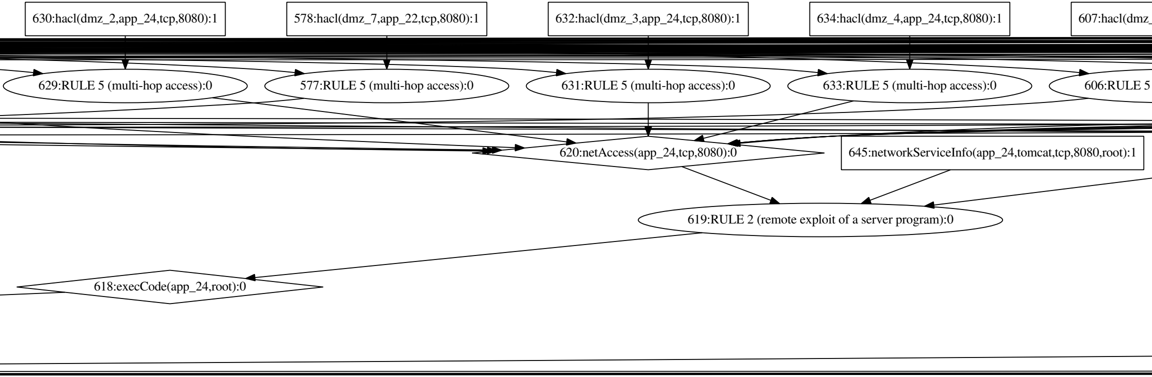

116:networkServiceInfo(dmz_9,httpd,tcp,80,root):1
1
112:RULE 2 (remote expl
574:hacl(dmz_5,app_22,tcp,8080):1
591:hacl(dmz_0,app_23,tcp,8080):1
593:hacl(dmz_1,app_23,tcp,8080):1
595:hacl(dmz_10,app_23,tcp,8080):1
573:RULE 5 (multi-hop access):0
590:RULE 5 (multi-hop access):0
592:RULE 5 (multi-hop access):0
594:RULE 5 (multi-hop access):0
ncat,remoteExploit,privEscalation):1
586:hacl(app_23,db_25,tcp,5432):1
585:RULE 5 (multi-hop access):0

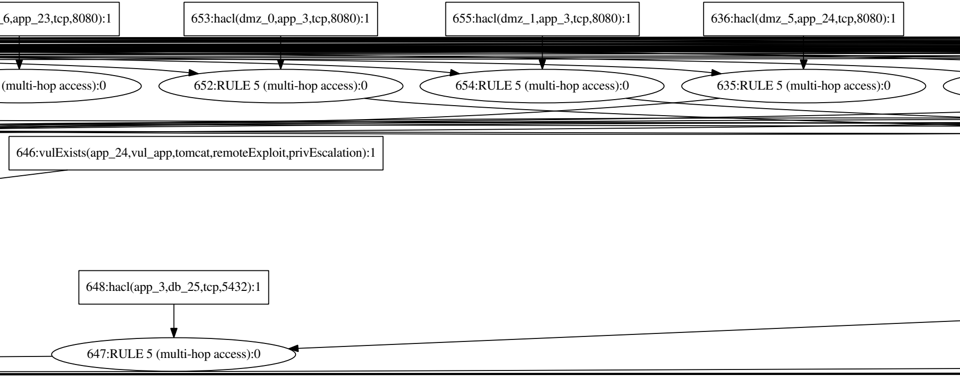

17:vulExists(dmz_9,vul_web,httpd,remoteExploit,privEscalation):1
loit of a server program):0
518:hacl(dmz_8,app_20,tcp,8080):1
597:hacl(dmz_11,app_23,tcp,8080):1
458:hacl(dmz_9,app_19,tcp,8080):1
111:execCode(dmz_9,root):0
517:RULE 5 (multi-hop access):0
596:RULE 5 (multi-hop access):0
457:RULE 5 (multi-hop access):0
488:RULE 5 (multi-hop
587:execCode(app_23,root):0

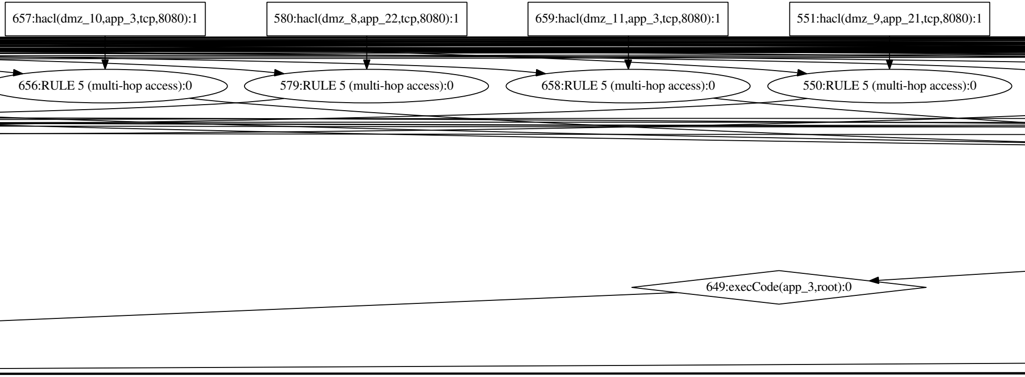

489:hacl(dmz_9,app_2,tcp,8080):1
599:hacl(dmz_2,app_23,tcp,8080):1
547:hacl(dmz_7,app_21,tcp,8080):1
601:hacl(dmz_3,app_23,tcp,8080):1
603:hacl(dmz_4,app_23,tcp,8080):1
access):0
598:RULE 5 (multi-hop access):0
546:RULE 5 (multi-hop access):0
600:RULE 5 (multi-hop access):0
602:RULE 5 (multi-hop access):0
589:netAccess(app_23,tcp,8080):0
614:networkServiceInfo(app_23,tom
588:RULE 2 (remote exploit of a server program):0

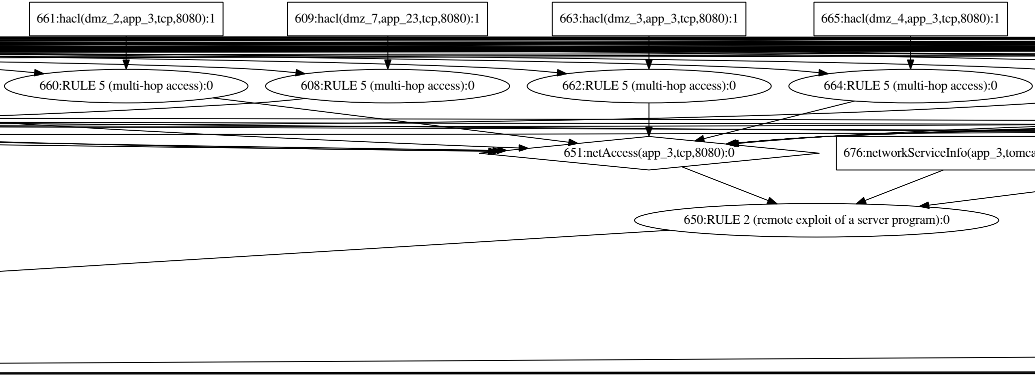

576:hacl(dmz_6,app_22,tcp,8080):1
622:hacl(dmz_0,app_24,tcp,8080):1
624:hacl(dmz_1,app_24,tcp,8080):1
605:hacl(dmz_5,app_23,tcp,8080):1
575:RULE 5 (multi-hop access):0
621:RULE 5 (multi-hop access):0
623:RULE 5 (multi-hop access):0
604:RULE 5 (multi-hop access):0
cat,tcp,8080,root):1
615:vulExists(app_23,vul_app,tomcat,remoteExploit,privEscalation):1
617:hacl(app_24,db_25,tcp,5432):1
616:RULE 5 (multi-hop access):0

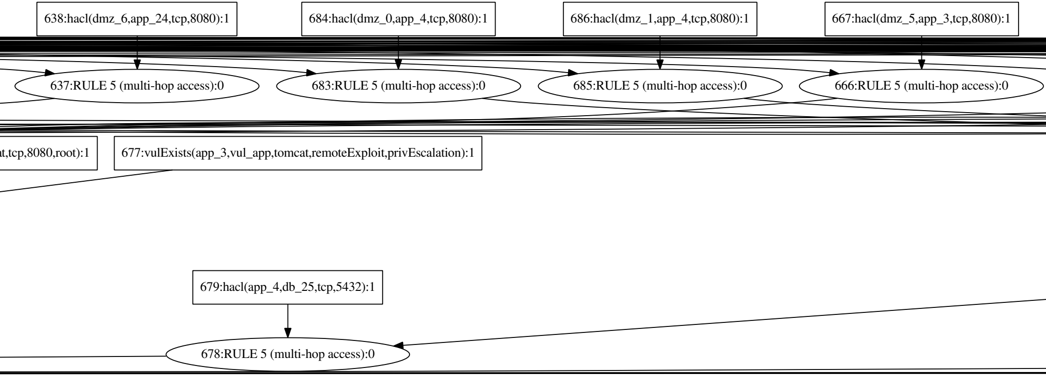

626:hacl(dmz_10,app_24,tcp,8080):1
549:hacl(dmz_8,app_21,tcp,8080):1
628:hacl(dmz_11,app_24,tcp,8080):1
520:hacl(dmz_9,app_20,tcp,8080):1
625:RULE 5 (multi-hop access):0
548:RULE 5 (multi-hop access):0
627:RULE 5 (multi-hop access):0
519:RULE 5 (multi-hop access):0

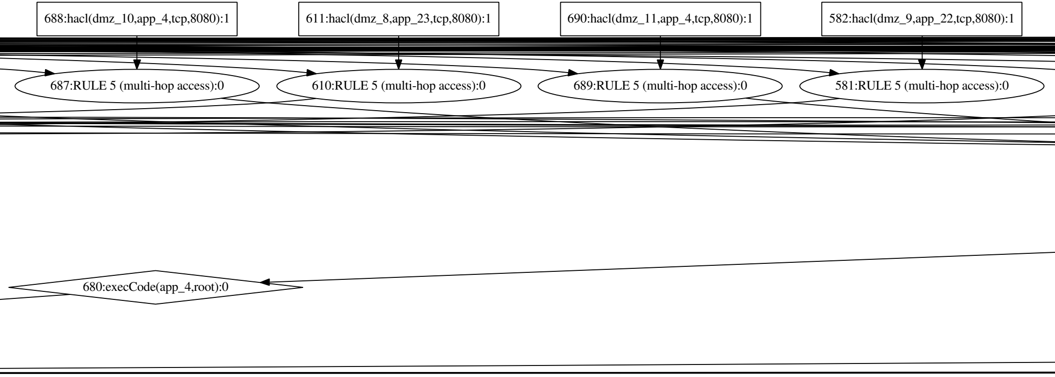

630:hacl(dmz_2,app_24,tcp,8080):1
578:hacl(dmz_7,app_22,tcp,8080):1
632:hacl(dmz_3,app_24,tcp,8080):1
634:hacl(dmz_4,app_24,tcp,8080):1
607:hacl(dmz_
629:RULE 5 (multi-hop access):0
577:RULE 5 (multi-hop access):0
631:RULE 5 (multi-hop access):0
633:RULE 5 (multi-hop access):0
606:RULE 5
620:netAccess(app_24,tcp,8080):0
645:networkServiceInfo(app_24,tomcat,tcp,8080,root):1
619:RULE 2 (remote exploit of a server program):0
618:execCode(app_24,root):0

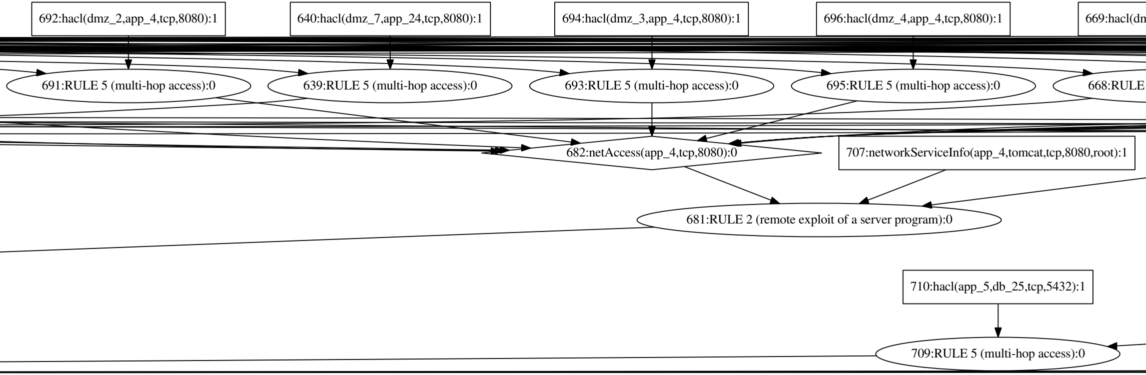

_6,app_23,tcp,8080):1
653:hacl(dmz_0,app_3,tcp,8080):1
655:hacl(dmz_1,app_3,tcp,8080):1
636:hacl(dmz_5,app_24,tcp,8080):1
(multi-hop access):0
652:RULE 5 (multi-hop access):0
654:RULE 5 (multi-hop access):0
635:RULE 5 (multi-hop access):0
646:vulExists(app_24,vul_app,tomcat,remoteExploit,privEscalation):1
648:hacl(app_3,db_25,tcp,5432):1
647:RULE 5 (multi-hop access):0

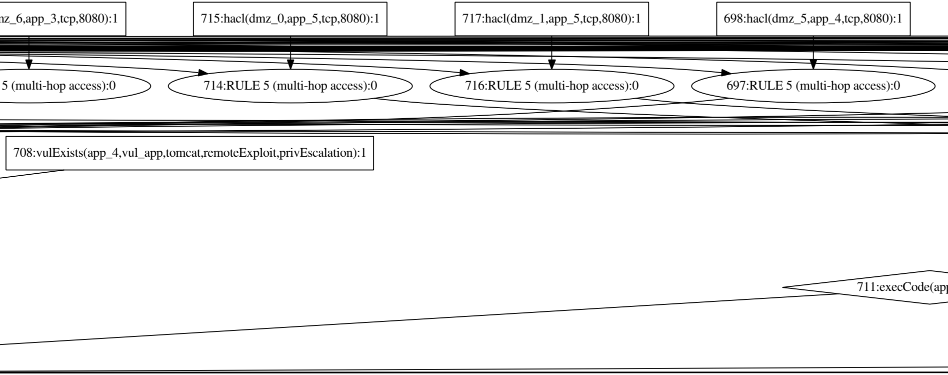

657:hacl(dmz_10,app_3,tcp,8080):1
580:hacl(dmz_8,app_22,tcp,8080):1
659:hacl(dmz_11,app_3,tcp,8080):1
551:hacl(dmz_9,app_21,tcp,8080):1
656:RULE 5 (multi-hop access):0
579:RULE 5 (multi-hop access):0
658:RULE 5 (multi-hop access):0
550:RULE 5 (multi-hop access):0
649:execCode(app_3,root):0

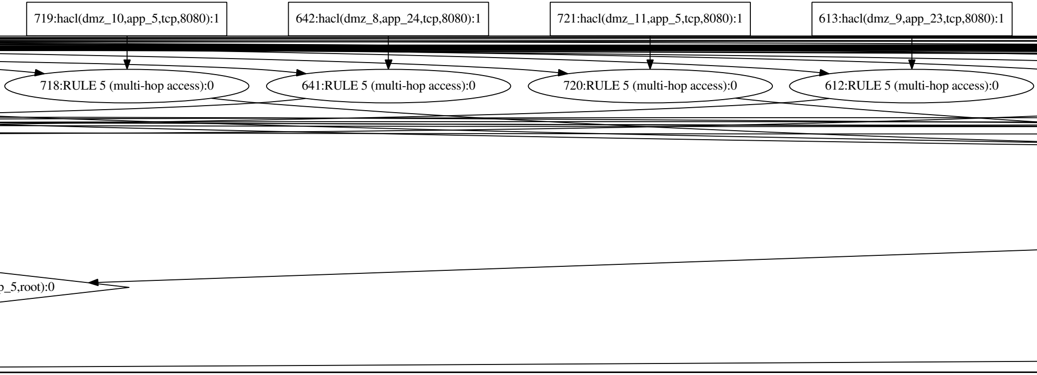

661:hacl(dmz_2,app_3,tcp,8080):1
609:hacl(dmz_7,app_23,tcp,8080):1
663:hacl(dmz_3,app_3,tcp,8080):1
665:hacl(dmz_4,app_3,tcp,8080):1
660:RULE 5 (multi-hop access):0
608:RULE 5 (multi-hop access):0
662:RULE 5 (multi-hop access):0
664:RULE 5 (multi-hop access):0
651:netAccess(app_3,tcp,8080):0
676:networkServiceInfo(app_3,tomca
650:RULE 2 (remote exploit of a server program):0

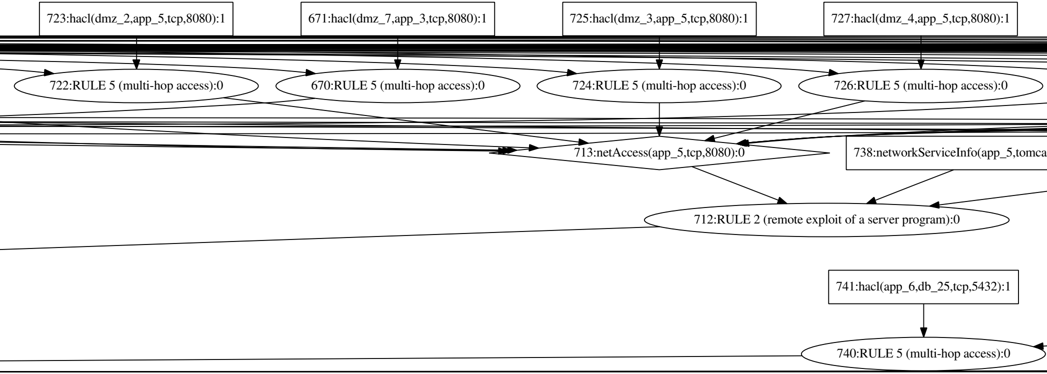

638:hacl(dmz_6,app_24,tcp,8080):1
684:hacl(dmz_0,app_4,tcp,8080):1
686:hacl(dmz_1,app_4,tcp,8080):1
667:hacl(dmz_5,app_3,tcp,8080):1
637:RULE 5 (multi-hop access):0
683:RULE 5 (multi-hop access):0
685:RULE 5 (multi-hop access):0
666:RULE 5 (multi-hop access):0
at,tcp,8080,root):1
677:vulExists(app_3,vul_app,tomcat,remoteExploit,privEscalation):1
679:hacl(app_4,db_25,tcp,5432):1
678:RULE 5 (multi-hop access):0

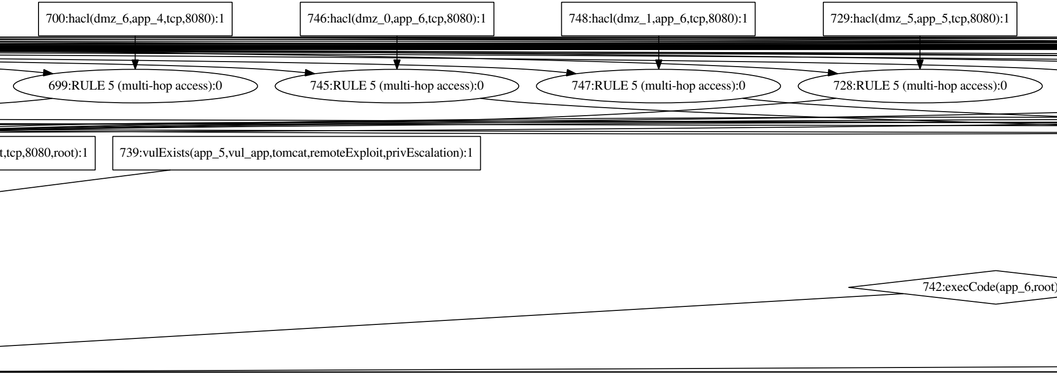

688:hacl(dmz_10,app_4,tcp,8080):1
611:hacl(dmz_8,app_23,tcp,8080):1
690:hacl(dmz_11,app_4,tcp,8080):1
582:hacl(dmz_9,app_22,tcp,8080):1
687:RULE 5 (multi-hop access):0
610:RULE 5 (multi-hop access):0
689:RULE 5 (multi-hop access):0
581:RULE 5 (multi-hop access):0
680:execCode(app_4,root):0

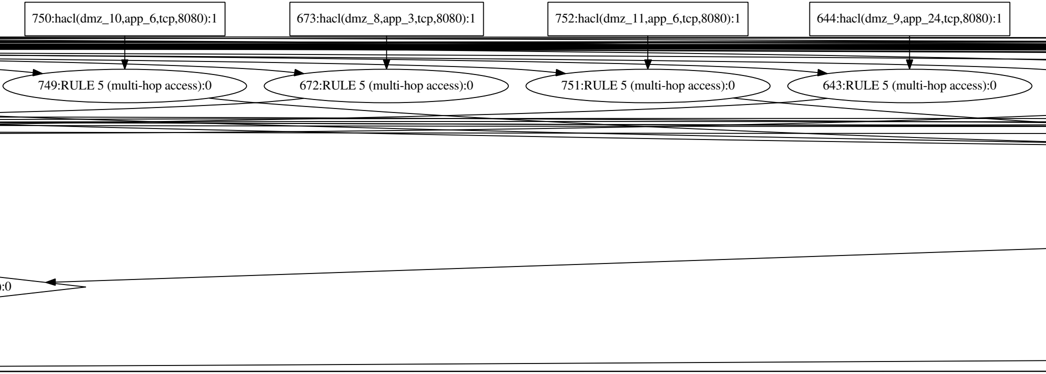

692:hacl(dmz_2,app_4,tcp,8080):1
640:hacl(dmz_7,app_24,tcp,8080):1
694:hacl(dmz_3,app_4,tcp,8080):1
696:hacl(dmz_4,app_4,tcp,8080):1
669:hacl(dm
691:RULE 5 (multi-hop access):0
639:RULE 5 (multi-hop access):0
693:RULE 5 (multi-hop access):0
695:RULE 5 (multi-hop access):0
668:RULE
682:netAccess(app_4,tcp,8080):0
707:networkServiceInfo(app_4,tomcat,tcp,8080,root):1
681:RULE 2 (remote exploit of a server program):0
710:hacl(app_5,db_25,tcp,5432):1
709:RULE 5 (multi-hop access):0

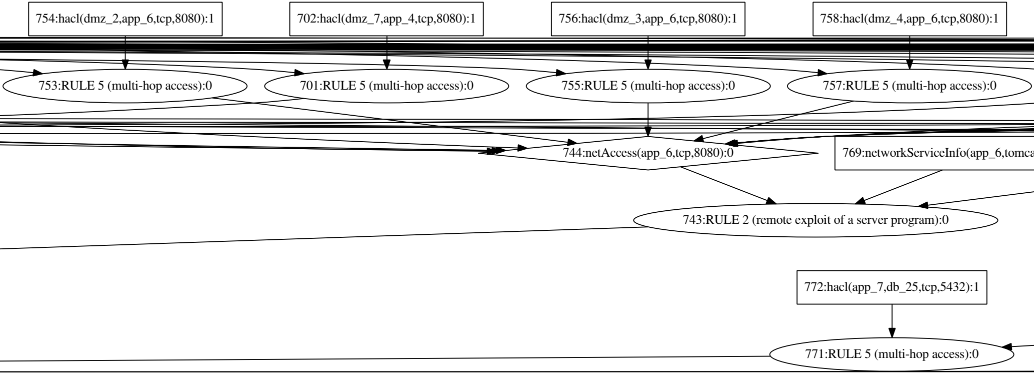

z_6,app_3,tcp,8080):1
715:hacl(dmz_0,app_5,tcp,8080):1
717:hacl(dmz_1,app_5,tcp,8080):1
698:hacl(dmz_5,app_4,tcp,8080):1
5 (multi-hop access):0
714:RULE 5 (multi-hop access):0
716:RULE 5 (multi-hop access):0
697:RULE 5 (multi-hop access):0
708:vulExists(app_4,vul_app,tomcat,remoteExploit,privEscalation):1
711:execCode(ap

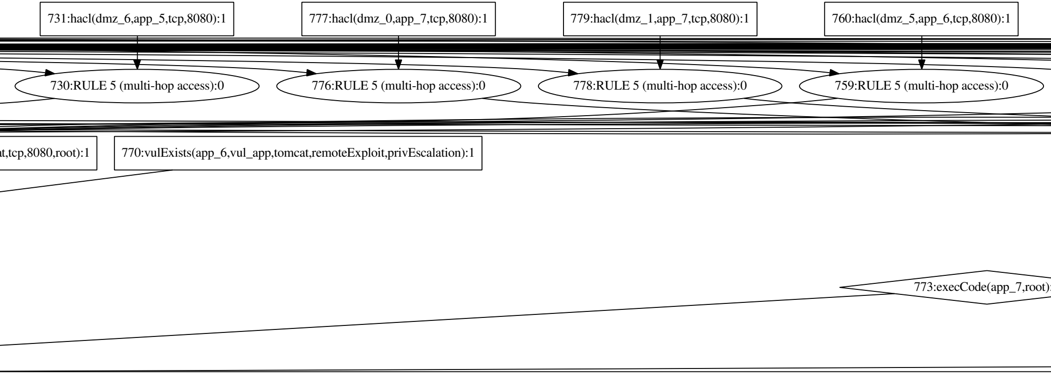

719:hacl(dmz_10,app_5,tcp,8080):1
642:hacl(dmz_8,app_24,tcp,8080):1
721:hacl(dmz_11,app_5,tcp,8080):1
613:hacl(dmz_9,app_23,tcp,8080):1
718:RULE 5 (multi-hop access):0
641:RULE 5 (multi-hop access):0
720:RULE 5 (multi-hop access):0
612:RULE 5 (multi-hop access):0
p_5,root):0

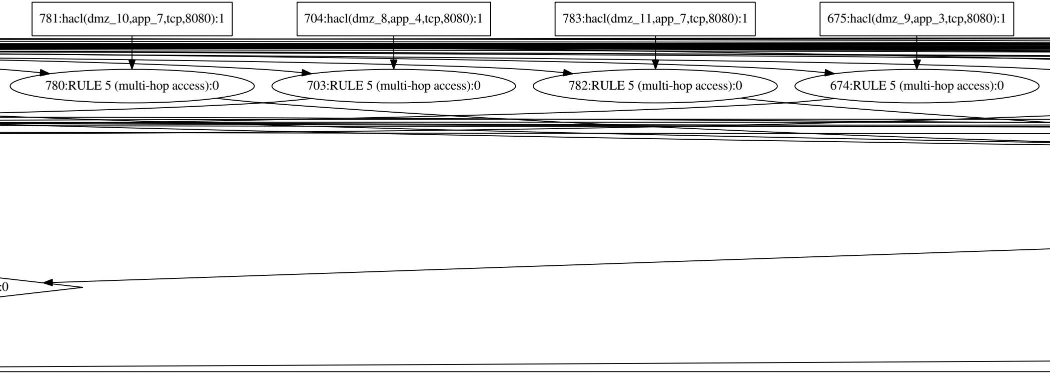

723:hacl(dmz_2,app_5,tcp,8080):1
671:hacl(dmz_7,app_3,tcp,8080):1
725:hacl(dmz_3,app_5,tcp,8080):1
727:hacl(dmz_4,app_5,tcp,8080):1
722:RULE 5 (multi-hop access):0
670:RULE 5 (multi-hop access):0
724:RULE 5 (multi-hop access):0
726:RULE 5 (multi-hop access):0
713:netAccess(app_5,tcp,8080):0
738:networkServiceInfo(app_5,tomca
712:RULE 2 (remote exploit of a server program):0
741:hacl(app_6,db_25,tcp,5432):1
740:RULE 5 (multi-hop access):0

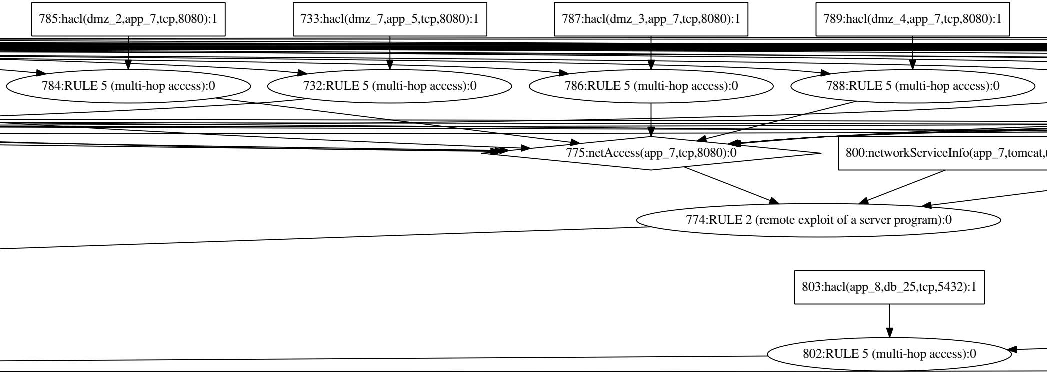

700:hacl(dmz_6,app_4,tcp,8080):1
746:hacl(dmz_0,app_6,tcp,8080):1
748:hacl(dmz_1,app_6,tcp,8080):1
729:hacl(dmz_5,app_5,tcp,8080):1
699:RULE 5 (multi-hop access):0
745:RULE 5 (multi-hop access):0
747:RULE 5 (multi-hop access):0
728:RULE 5 (multi-hop access):0
t,tcp,8080,root):1
739:vulExists(app_5,vul_app,tomcat,remoteExploit,privEscalation):1
742:execCode(app_6,root)

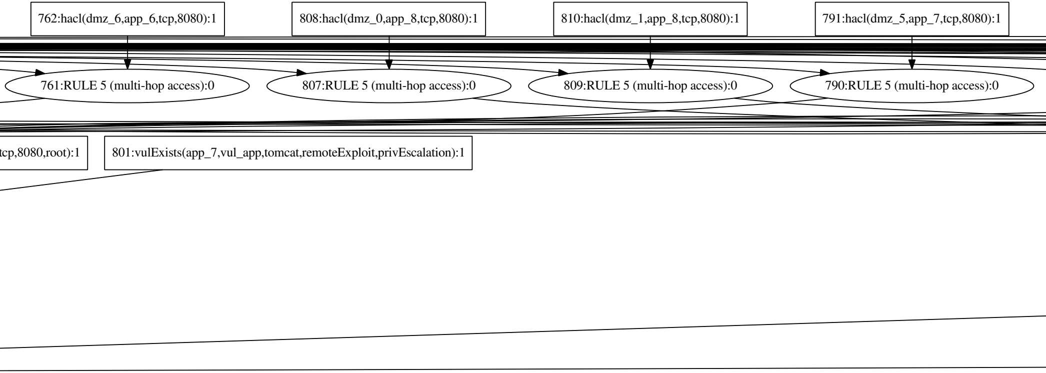

750:hacl(dmz_10,app_6,tcp,8080):1
673:hacl(dmz_8,app_3,tcp,8080):1
752:hacl(dmz_11,app_6,tcp,8080):1
644:hacl(dmz_9,app_24,tcp,8080):1
749:RULE 5 (multi-hop access):0
672:RULE 5 (multi-hop access):0
751:RULE 5 (multi-hop access):0
643:RULE 5 (multi-hop access):0
:0

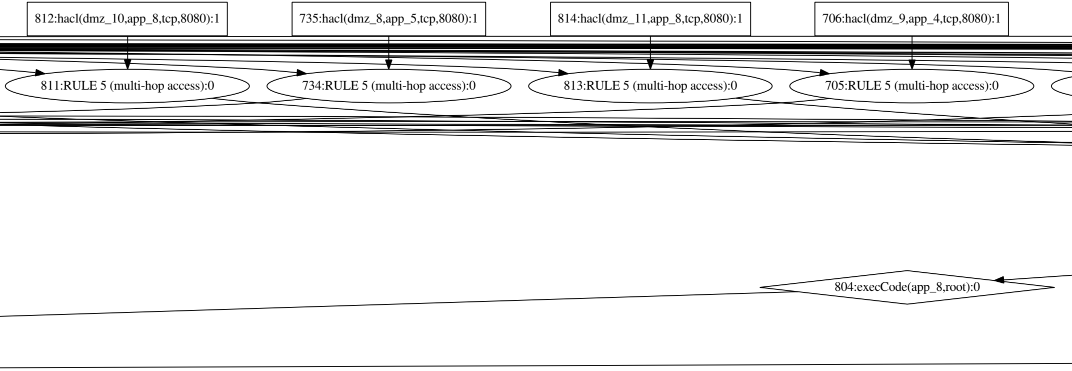

754:hacl(dmz_2,app_6,tcp,8080):1
702:hacl(dmz_7,app_4,tcp,8080):1
756:hacl(dmz_3,app_6,tcp,8080):1
758:hacl(dmz_4,app_6,tcp,8080):1
753:RULE 5 (multi-hop access):0
701:RULE 5 (multi-hop access):0
755:RULE 5 (multi-hop access):0
757:RULE 5 (multi-hop access):0
744:netAccess(app_6,tcp,8080):0
769:networkServiceInfo(app_6,tomca
743:RULE 2 (remote exploit of a server program):0
772:hacl(app_7,db_25,tcp,5432):1
771:RULE 5 (multi-hop access):0

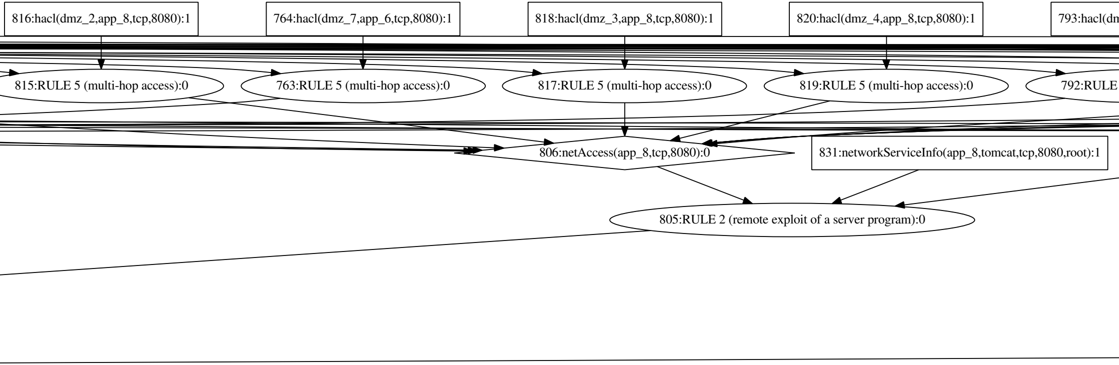

731:hacl(dmz_6,app_5,tcp,8080):1
777:hacl(dmz_0,app_7,tcp,8080):1
779:hacl(dmz_1,app_7,tcp,8080):1
760:hacl(dmz_5,app_6,tcp,8080):1
730:RULE 5 (multi-hop access):0
776:RULE 5 (multi-hop access):0
778:RULE 5 (multi-hop access):0
759:RULE 5 (multi-hop access):0
t,tcp,8080,root):1
770:vulExists(app_6,vul_app,tomcat,remoteExploit,privEscalation):1
773:execCode(app_7,root):

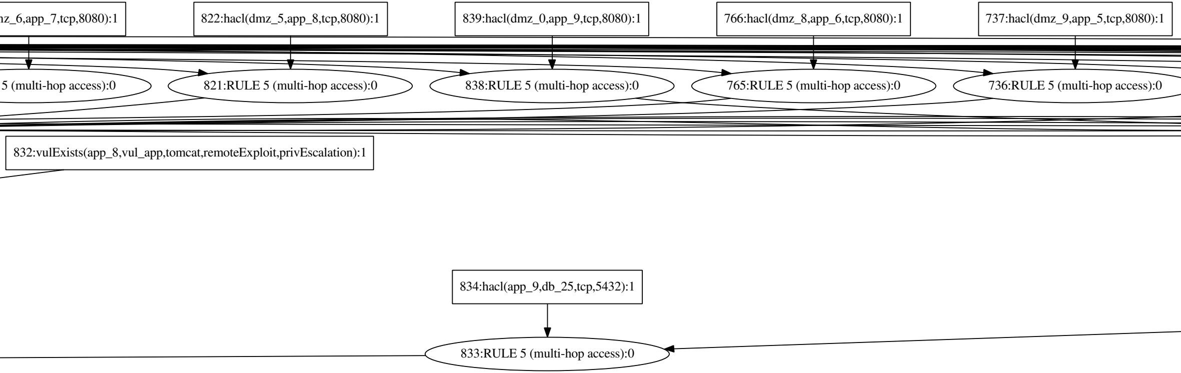

781:hacl(dmz_10,app_7,tcp,8080):1
704:hacl(dmz_8,app_4,tcp,8080):1
783:hacl(dmz_11,app_7,tcp,8080):1
675:hacl(dmz_9,app_3,tcp,8080):1
780:RULE 5 (multi-hop access):0
703:RULE 5 (multi-hop access):0
782:RULE 5 (multi-hop access):0
674:RULE 5 (multi-hop access):0
:0

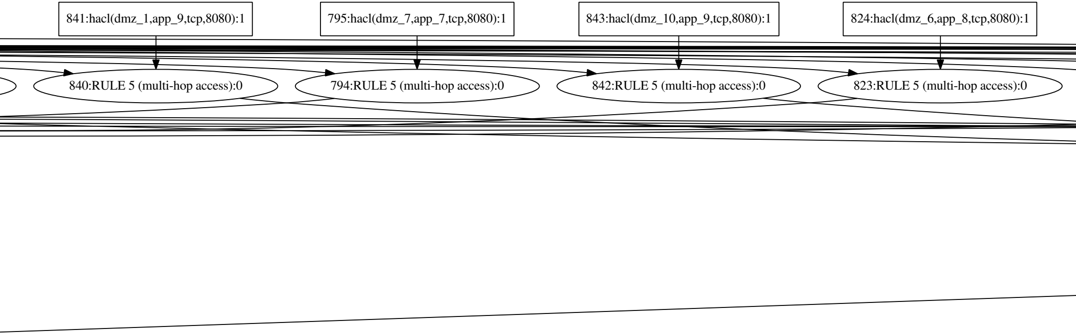

785:hacl(dmz_2,app_7,tcp,8080):1
733:hacl(dmz_7,app_5,tcp,8080):1
787:hacl(dmz_3,app_7,tcp,8080):1
789:hacl(dmz_4,app_7,tcp,8080):1
784:RULE 5 (multi-hop access):0
732:RULE 5 (multi-hop access):0
786:RULE 5 (multi-hop access):0
788:RULE 5 (multi-hop access):0
775:netAccess(app_7,tcp,8080):0
800:networkServiceInfo(app_7,tomcat,t
774:RULE 2 (remote exploit of a server program):0
803:hacl(app_8,db_25,tcp,5432):1
802:RULE 5 (multi-hop access):0

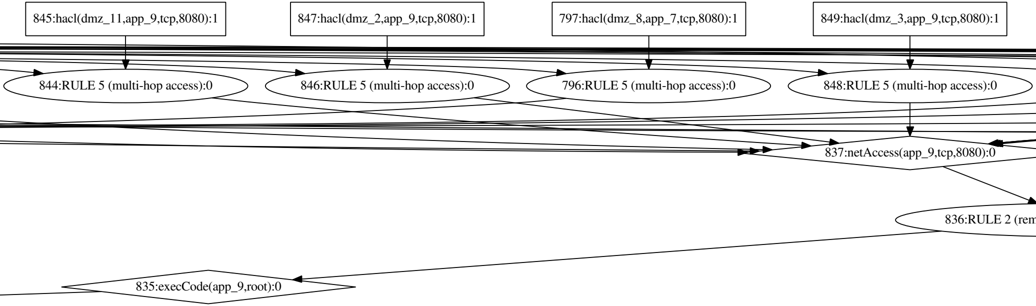

762:hacl(dmz_6,app_6,tcp,8080):1
808:hacl(dmz_0,app_8,tcp,8080):1
810:hacl(dmz_1,app_8,tcp,8080):1
791:hacl(dmz_5,app_7,tcp,8080):1
761:RULE 5 (multi-hop access):0
807:RULE 5 (multi-hop access):0
809:RULE 5 (multi-hop access):0
790:RULE 5 (multi-hop access):0
tcp,8080,root):1
801:vulExists(app_7,vul_app,tomcat,remoteExploit,privEscalation):1

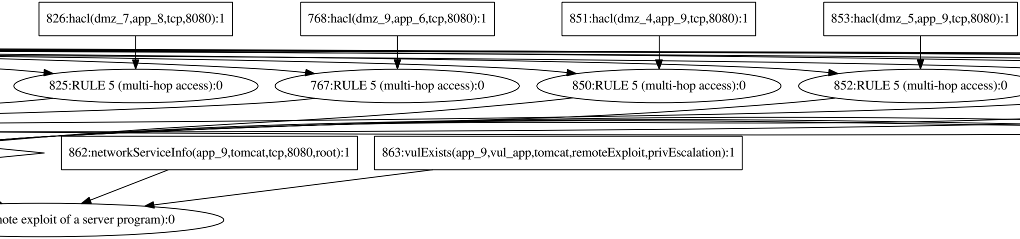

812:hacl(dmz_10,app_8,tcp,8080):1
735:hacl(dmz_8,app_5,tcp,8080):1
814:hacl(dmz_11,app_8,tcp,8080):1
706:hacl(dmz_9,app_4,tcp,8080):1
811:RULE 5 (multi-hop access):0
734:RULE 5 (multi-hop access):0
813:RULE 5 (multi-hop access):0
705:RULE 5 (multi-hop access):0
804:execCode(app_8,root):0

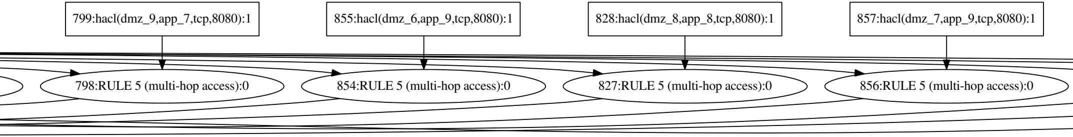

816:hacl(dmz_2,app_8,tcp,8080):1
764:hacl(dmz_7,app_6,tcp,8080):1
818:hacl(dmz_3,app_8,tcp,8080):1
820:hacl(dmz_4,app_8,tcp,8080):1
793:hacl(dm
815:RULE 5 (multi-hop access):0
763:RULE 5 (multi-hop access):0
817:RULE 5 (multi-hop access):0
819:RULE 5 (multi-hop access):0
792:RULE
806:netAccess(app_8,tcp,8080):0
831:networkServiceInfo(app_8,tomcat,tcp,8080,root):1
805:RULE 2 (remote exploit of a server program):0

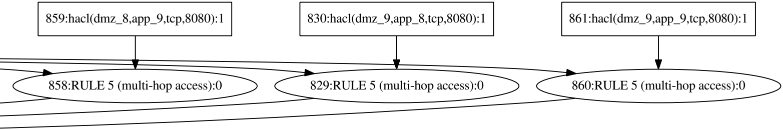

nz_6,app_7,tcp,8080):1
822:hacl(dmz_5,app_8,tcp,8080):1
839:hacl(dmz_0,app_9,tcp,8080):1
766:hacl(dmz_8,app_6,tcp,8080):1
737:hacl(dmz_9,app_5,tcp,8080):1
5 (multi-hop access):0
821:RULE 5 (multi-hop access):0
838:RULE 5 (multi-hop access):0
765:RULE 5 (multi-hop access):0
736:RULE 5 (multi-hop access):0
832:vulExists(app_8,vul_app,tomcat,remoteExploit,privEscalation):1
834:hacl(app_9,db_25,tcp,5432):1
833:RULE 5 (multi-hop access):0


841:hacl(dmz_1,app_9,tcp,8080):1
795:hacl(dmz_7,app_7,tcp,8080):1
843:hacl(dmz_10,app_9,tcp,8080):1
824:hacl(dmz_6,app_8,tcp,8080):1
840:RULE 5 (multi-hop access):0
794:RULE 5 (multi-hop access):0
842:RULE 5 (multi-hop access):0
823:RULE 5 (multi-hop access):0


845:hacl(dmz_11,app_9,tcp,8080):1
847:hacl(dmz_2,app_9,tcp,8080):1
797:hacl(dmz_8,app_7,tcp,8080):1
849:hacl(dmz_3,app_9,tcp,8080):1
844:RULE 5 (multi-hop access):0
846:RULE 5 (multi-hop access):0
796:RULE 5 (multi-hop access):0
848:RULE 5 (multi-hop access):0
837:netAccess(app_9,tcp,8080):0
836:RULE 2 (rem
835:execCode(app_9,root):0


826:hacl(dmz_7,app_8,tcp,8080):1
768:hacl(dmz_9,app_6,tcp,8080):1
851:hacl(dmz_4,app_9,tcp,8080):1
853:hacl(dmz_5,app_9,tcp,8080):1
825:RULE 5 (multi-hop access):0
767:RULE 5 (multi-hop access):0
850:RULE 5 (multi-hop access):0
852:RULE 5 (multi-hop access):0
862:networkServiceInfo(app_9,tomcat,tcp,8080,root):1
863:vulExists(app_9,vul_app,tomcat,remoteExploit,privEscalation):1
note exploit of a server program):0


799:hacl(dmz_9,app_7,tcp,8080):1
855:hacl(dmz_6,app_9,tcp,8080):1
828:hacl(dmz_8,app_8,tcp,8080):1
857:hacl(dmz_7,app_9,tcp,8080):1
798:RULE 5 (multi-hop access):0
854:RULE 5 (multi-hop access):0
827:RULE 5 (multi-hop access):0
856:RULE 5 (multi-hop access):0


859:hacl(dmz_8,app_9,tcp,8080):1
830:hacl(dmz_9,app_8,tcp,8080):1
861:hacl(dmz_9,app_9,tcp,8080):1
858:RULE 5 (multi-hop access):0
829:RULE 5 (multi-hop access):0
860:RULE 5 (multi-hop access):0